

Hawking Radiation Without Fields

Phase Theory: Admissibility Leakage at Phase Saturation Boundaries

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Abstract

We derive Hawking radiation in Phase Theory without invoking quantum fields, vacuum fluctuations, particle-antiparticle pairs, or curved-spacetime field quantization. Radiation emerges as a consequence of admissibility leakage at phase saturation boundaries, where global phase consistency permits reconfiguration pathways that project as outgoing energy flux. The Hawking

spectrum arises from phase entropy gradients and admissibility balance, not from particle creation in a vacuum. We further show that the information paradox dissolves under phase-native description, that the trans-Planckian problem does not arise, that the firewall paradox is avoidable without postulating exotic boundary conditions, and that the Page curve is recovered through admissibility topology rather than quantum extremal surfaces. This formulation reproduces all known Hawking radiation results while eliminating their ontological paradoxes. We derive the Bekenstein-Hawking entropy from first principles within Phase Theory, establish the phase-native Hawking temperature formula, and enumerate five classes of falsifiable predictions distinguishing Phase Theory from quantum field theory on curved spacetime.

Contents

1. Why Hawking Radiation Is Ontologically Fragile
2. What Must Be Explained
3. Phase Theory Axioms Relevant to This Derivation
4. Phase Saturation Boundaries Revisited
5. The Global Consistency Enforcement Principle
6. The Admissibility Density Functional
7. Phase Saturation and Its Topological Structure
8. Definition and Formal Derivation of Admissibility Leakage
9. Radiation Without Creation — The Mechanism
10. Why the Radiation Is Thermal — Statistical Mechanics of Admissibility
11. Temperature from the Phase Entropy Gradient — Formal Derivation
12. Derivation of the Bekenstein-Hawking Entropy from Phase Principles
13. Universality of the Radiation — Why All Species Radiate Identically

- 14.** Energy Conservation — Formal Proof
- 15.** No Trans-Planckian Problem — Why Saturation Eliminates It
- 16.** Observer Independence of the Radiation
- 17.** The Unruh Effect Reinterpreted
- 18.** Late-Time Evaporation and the Final State
- 19.** The Information Paradox Dissolved
- 20.** The Page Curve Without Quantum Extremal Surfaces
- 21.** Firewall Resolution
- 22.** Complementarity Without Observers
- 23.** Near-Extremal Phase-Saturated Objects
- 24.** Rotating Phase-Saturated Objects (Kerr Analogue)
- 25.** Charged Phase-Saturated Objects (Reissner-Nordström Analogue)
- 26.** Phase-Native Derivation of the Stefan-Boltzmann Luminosity
- 27.** Phase Topology and Greybody Factors
- 28.** Black Hole Mining and Stimulated Leakage
- 29.** Comparison with Quantum Field Theory on Curved Spacetime
- 30.** Comparison with String Theory Black Holes
- 31.** Comparison with Loop Quantum Gravity Black Holes
- 32.** Comparison with Fuzzball and Firewall Proposals
- 33.** Primordial Phase-Saturated Objects and Cosmological Implications
- 34.** Analog Gravity and Laboratory Tests
- 35.** Objections and Responses
- 36.** Five Classes of Falsifiable Predictions
- 37.** Relation to the Thermodynamics of Spacetime
- 38.** Holography and AdS/CFT from Phase Theory
- 39.** Summary and Outlook

References

1. Why Hawking Radiation Is Ontologically Fragile

The discovery of black hole radiation by Hawking in 1974 stands as one of the most celebrated results in theoretical physics. By combining the semiclassical formalism of quantum fields propagating on a fixed curved background spacetime, Hawking showed that black holes radiate with a thermal spectrum at temperature proportional to the surface gravity of the horizon. The result brought together general relativity, quantum field theory, and thermodynamics in a single formula of extraordinary compactness. Yet the conceptual foundations of the derivation have remained contested for over five decades. This paper argues that the standard derivation rests on three pillars, each of which is heuristic, observer-dependent, or ontologically ambiguous, and that a derivation free from these defects is both possible and necessary.

The first pillar is quantum fields on curved spacetime. The formalism treats the spacetime metric as a classical background and quantizes matter fields propagating thereon. This semiclassical approximation has no known regime of strict validity near a horizon: the very region from which radiation is supposed to originate is one in which quantum gravitational corrections should, in principle, be important. The Bogoliubov transformation relating in-modes to out-modes depends on the global causal structure of the spacetime, including the existence of the event horizon as a global object. This dependence on the global structure is already a warning sign: the physical process of radiation cannot wait until the end of the universe to determine whether a horizon has formed.

The second pillar is the vacuum fluctuation narrative. In the popular and even in many technical treatments, Hawking radiation is described as arising from virtual particle-antiparticle pairs spontaneously created near the horizon: one member falls in, the other escapes. This narrative, while evocative, has no precise mathematical content. The notion of a particle is frame-dependent in curved spacetime; the vacuum of one observer is not the vacuum of another. The pair-creation metaphor

does not survive examination: virtual particles are internal lines in Feynman diagrams, not physical entities, and the horizon is not a membrane that can physically intercept one member of a pair. Hawking himself acknowledged that this picture is merely a heuristic.

The third pillar is the particle-creation framework. The mode decomposition of the field into in and out sectors, the Bogoliubov transformation relating them, and the identification of particle number as the relevant physical observable all depend on an asymptotic flat region where the notion of a particle is well-defined. In the absence of such an asymptotic region — as in cosmological settings or in the final stages of black hole evaporation — the particle-creation framework breaks down. The description is inherently ambiguous for observers who do not share the asymptotic structure.

The historical arc of these difficulties is instructive. Hawking's 1974 paper established the result. His 1976 paper showed that evaporation implies information loss, launching the information paradox. Unruh (1976) showed that an accelerating observer in flat spacetime perceives a thermal bath, adding observer-dependence to the conceptual landscape. The next three decades produced a proliferation of approaches — black hole complementarity, string-theoretic microstate counting, the fuzzball proposal — without resolving the foundational issues. The AMPS firewall argument of Almheiri, Marolf, Polchinski, and Sully (2013) brought the tensions to a head: they showed that unitarity, the equivalence principle, and quantum field theory cannot simultaneously hold at the horizon. The firewall is not a solution; it is a diagnosis. It tells us that at least one of the standard assumptions must break down at the horizon, but it does not tell us which one, nor does it provide a replacement framework.

Phase Theory demands a derivation that works without fields, vacua, or particle-creation metaphors. The reason is not mere aesthetic preference. Phase Theory posits that these concepts are emergent — effective descriptions valid in regimes far from saturation. Near a saturation boundary, the emergent ontology of fields and particles breaks down not because of technical computational difficulties but because the underlying reality has a different structure. A derivation of Hawking radiation that relies on field quantization is, from the perspective of Phase Theory,

analogous to deriving the boiling point of water from a model that treats water as a continuous fluid: correct in a regime, wrong in principle, and unable to answer questions about the microscopic mechanism. The derivation presented in this paper is constructed entirely from the axioms of Phase Theory, without reference to quantum fields, vacuum states, particle creation, or Bogoliubov coefficients.

The structure of the argument is as follows. Black holes, in Phase Theory, are phase-saturated objects: regions in which the admissibility density of the phase substrate has reached its maximum value. The saturation boundary is a topological interface, not a coordinate artifact. Global consistency enforcement — the mechanism that maintains the global admissibility of the phase configuration — requires that phase structure leak outward across this boundary. This leakage is what appears, in the emergent spacetime description, as Hawking radiation. The thermal character of the radiation, its temperature, its entropy, and its information content all follow from the topology and statistical mechanics of the leakage pathways, without any reference to quantum fields.

2. What Must Be Explained

Before presenting the Phase Theory derivation, it is essential to enumerate precisely what any acceptable derivation of Hawking radiation must account for. This section provides that enumeration, with commentary on why each item represents a genuine explanatory demand rather than merely a numerical coincidence to be reproduced.

(a) Steady thermal radiation. An isolated black hole in an otherwise empty universe should emit radiation continuously, at a steady rate, with a spectrum that is, to leading order, exactly Planckian. This is not approximate thermality but exact thermality in the idealized case of a non-rotating, uncharged black hole. The derivation must explain why the process is steady, why it does not require an external trigger, and why the spectrum is Planckian rather than some other distribution.

(b) Temperature inversely proportional to mass. The Hawking temperature is:

$$T_H = \hbar c^3 / (8\pi G M k_B)$$

(49.1)

A more massive black hole is colder. This inverse relationship between mass and temperature is opposite to the behavior of ordinary thermodynamic systems. The derivation must explain not only the numerical coefficient but the functional form and the physical reason for the inverse relationship. In particular, it must explain why larger objects radiate at lower temperatures without invoking the surface gravity of the event horizon (which is a GR-specific concept unavailable in a field-free derivation).

(c) Universality across all species. The radiation is thermal and democratic: all species of particles — photons, gravitons, fermions, scalars — are emitted at the same temperature. The derivation must account for this universality. In standard QFT derivations, it follows from the fact that all fields couple to the same curved background geometry. In Phase Theory, a different explanation is required, since there is no background geometry at the fundamental level.

(d) Energy loss and evaporation. As radiation is emitted, the black hole loses mass. The evaporation rate must be derived consistently with energy conservation. The derivation must show that energy lost by the black hole exactly equals energy carried away by radiation, without negative-energy particles, exotic matter, or fine-tuned cancellations. The timescale for complete evaporation must be derived and must scale as $t_{\text{evap}} \sim M_0^3$.

(e) Entropy scaling as area. The Bekenstein-Hawking entropy is:

$$S_{BH} = k_B A / (4l_P^2)$$

(49.2)

where A is the area of the event horizon and $l_P = \sqrt{\hbar G/c^3}$ is the Planck length. The derivation must reproduce this formula from first principles. Crucially, it must explain why the entropy scales with area rather than volume — a result deeply

puzzling from the perspective of conventional statistical mechanics, where entropy is an extensive bulk quantity.

(f) Information content of the outgoing radiation. If Hawking radiation carries no information about the initial state, then black hole evaporation violates unitarity. If it does carry information, the mechanism must be specified. The derivation must make a definite claim about information content and provide a mechanism — not merely an assertion — by which information is preserved. This is arguably the most demanding item on the list, since no existing framework provides a fully satisfactory answer.

Phase Theory, as developed in this paper, provides accounts of all six items. The derivation is self-contained from the axioms established in Papers 1 and 47–48, and does not assume any result from quantum field theory, string theory, or loop quantum gravity. The matching of numerical results — including the coefficient in (49.1) and (49.2) — is a structural consequence of the formalism, not a post-hoc adjustment of parameters.

3. Phase Theory Axioms Relevant to This Derivation

Phase Theory, developed over Papers 1 through 48 of this series, is a unified theoretical framework in which phase is posited as the sole fundamental constituent of physical reality. Space, time, matter, light, and geometry are emergent structures arising from the global organization, topology, and consistency constraints of a single phase-bearing substrate. This section states and elaborates the four axioms of Phase Theory most directly relevant to the derivation of Hawking radiation.

Axiom $\Phi 1$ (Primacy of Phase)

The fundamental ontological substrate is a phase field ϕ , described by a map $\varphi: M \rightarrow S^1$ from a topological space M (the pre-geometric manifold) to the circle S^1 .

All physical quantities are functionals of φ . There exist no fundamental fields, particles, or geometric structures that are not reducible to phase organization.

Axiom $\Phi 1$ was established in Paper 1 of this series and refined in Papers 12 and 23. Its radical character deserves emphasis. The claim is not that fields and particles are described by phase functions — which would be a modest technical claim — but that fields and particles have no existence except as particular patterns of global phase organization. The electron, the photon, the graviton: all are phase defect structures, patterns of winding and coherence in the field φ .

Axiom $\Phi 2$ (Global Consistency / Admissibility)

The realized configuration space of Phase Theory is restricted to the admissible set:

$$Conf_\phi = \{ \varphi \in Maps(M, S^1) : \varphi \text{ satisfies the global topological consistency constraints } C_1, ..., C_n \}$$

(49.3)

The constraints C_i include global winding number conservation, phase coherence across connected components, and saturation bounds on local phase variation density. Physical evolution is a path in $Conf_\phi$; configurations outside $Conf_\phi$ are not physically realized.

Axiom $\Phi 2$ is the most structurally rich axiom in Phase Theory. It is the source of quantum-like phenomena (the discreteness of admissible transitions), relativistic phenomena (the causal structure of configuration space), and, as shown in this paper, thermal radiation from saturated regions. The admissibility constraints are global: a configuration is admissible only if its entire phase structure simultaneously satisfies all consistency conditions. This is why phenomena like entanglement and,

as argued here, Hawking radiation are intrinsically non-local in configuration space while remaining causal in the emergent spacetime sense.

Axiom $\Phi 3$ (Emergence of Physical Structure)

Spacetime geometry, particle species, gauge fields, and their interactions are effective descriptions of large-scale phase organization. Specifically: (i) the emergent metric tensor is determined by the two-point correlations of φ ; (ii) matter fields correspond to topologically non-trivial defect structures in φ ; (iii) gauge symmetries correspond to continuous families of admissibility-preserving phase reparametrizations.

Axiom $\Phi 3$ was partially established in Papers 7, 14, and 31–35, where the emergent Riemannian metric and the Standard Model gauge structure were derived from phase organization. It licenses the use of spacetime language as an effective description while forbidding the confusion of that effective description with fundamental ontology. In particular, the event horizon of general relativity is, under $\Phi 3$, an effective description of the saturation boundary — not a fundamental geometric object.

Axiom $\Phi 4$ (Phase Entropy)

The phase entropy of a region R is:

$$S_{\Phi}(R) = k_B \ln |\Omega_{\Phi}(R)|$$

(49.4)

where $\Omega_{\Phi}(R)$ is the number of admissible microstates of the phase field in region R consistent with the macroscopic boundary conditions. The phase temperature is $T_{\Phi}^{-1} = \partial S_{\Phi} / \partial E_{\Phi}$, where E_{Φ} is the total phase energy (the emergent mass-energy).

Axiom $\Phi 4$ connects the Phase Theory formalism to thermodynamics in a direct and natural way. Unlike in standard QFT derivations, where the black hole temperature must be extracted via Bogoliubov transformations, in Phase Theory the temperature is the derivative of the entropy with respect to energy — the standard thermodynamic definition. This means that the entire apparatus of thermodynamics is available to Phase Theory without any auxiliary assumptions. The derivation in Section 11 exploits this directly.

Together, these four axioms provide a complete and self-consistent foundation for the derivation presented in the following sections. No additional assumptions are introduced that are not consequences of $\Phi 1$ – $\Phi 4$. Where mathematical constructs are introduced (the admissibility density functional, the GCE operator, leakage pathways), they are explicitly defined as functionals of φ satisfying the admissibility constraints.

4. Phase Saturation Boundaries Revisited

The concept of phase saturation was introduced in Papers 47 and 48. This section provides a more detailed exposition for the purposes of the current derivation. Phase saturation is the condition in which the admissibility density of the phase substrate has reached its absolute maximum at every interior point of a region. Understanding this condition precisely is essential to the argument for admissibility leakage.

Definition 49.1 (Phase Saturation)

A region $R \subset M$ is *phase-saturated* if and only if the admissibility density $\rho_{adm}(x) = \rho_{max}$ for all $x \in Int(R)$, where $\rho_{max} = c^3/(4\hbar G k_B)$ is the maximum admissibility density consistent with Axiom $\Phi 2$.

The value ρ_{max} is not an independently posited parameter. It follows from the consistency constraints of Axiom $\Phi 2$: specifically, from the requirement that the

local winding number density cannot exceed one topological unit per Planck area. The Planck area $l_p^2 = \hbar G/c^3$ appears here as the fundamental granularity of the admissible configuration space, derived from the phase substrate structure rather than assumed.

The significance of saturation is profound. An interior point of a saturated region cannot carry any additional phase distinction — it is already in the most informationally dense admissible state possible. This means that perturbations of the interior phase structure are forbidden: no new phase gradient can be inscribed, no new topological feature can be created, and no additional information can be stored. The interior is, in a precise sense, informationally frozen.

The saturation radius R_{sat} is defined as the radius of the smallest sphere enclosing the saturated region. For a spherically symmetric saturated object of total phase energy $E_\phi = Mc^2$, the saturation condition requires that the total admissibility within the sphere exactly equals the capacity of that sphere:

$$R_{sat} = 2GM/c^2 \tag{49.5}$$

This is exactly the Schwarzschild radius. This is not a coincidence. In the effective spacetime description generated by $\Phi 3$, the saturation boundary maps to the event horizon. But the saturation boundary is not defined by the causal structure of the emergent spacetime — it is defined by the topology of the admissible configuration space. The event horizon is a derivative concept; the saturation boundary is primary. This distinction has direct consequences: the saturation boundary is not a coordinate artifact (it does not depend on the choice of emergent time coordinate), it is not observer-dependent, and it is not ambiguous in the way that the global event horizon of GR is ambiguous (which requires knowledge of the entire future of the spacetime to locate).

How does saturation arise dynamically? When a sufficient density of phase complexity concentrates in a bounded region, the local admissibility density rises toward ρ_{max} . This is the Phase Theory analogue of gravitational collapse. As the

admissibility density approaches ρ_{max} , the phase substrate becomes increasingly rigid: perturbations that would require new topological features are rejected by the admissibility constraints. The final state of this process is a saturated region bounded by a sharp topological interface — the saturation boundary $\partial\Sigma_{sat}$.

The saturation boundary is not a smooth manifold in general. Its topological character is determined by the winding structure of the global phase field, which may be non-trivially knotted, linked, or braided at the boundary. However, for the leading-order derivation of Hawking radiation, the relevant quantity is the total winding deficit (Section 7), which is determined by the total admissibility budget M rather than by the detailed boundary topology. The detailed topology produces subleading corrections discussed in later sections.

5. The Global Consistency Enforcement

Principle

Global Consistency Enforcement (GCE) is the dynamical mechanism through which Phase Theory maintains Axiom $\Phi 2$. It is the most important single mechanism in Phase Theory, responsible for quantum entanglement correlations, the emergence of causal structure, the stability of defect structures (particles), and — as argued in this paper — Hawking-like radiation from phase-saturated objects.

Definition 49.2 (GCE Operator)

The Global Consistency Enforcement operator $\Gamma[\varphi]$ is the projection operator on the space of phase configurations that maps any configuration to its nearest admissible configuration in the $L^2(M, S^1)$ metric:

$$\Gamma[\varphi] = \operatorname{argmin}_{\psi \in \text{Conf}\Phi} \|\varphi - \psi\|_{L^2}$$

(49.6)

Physical evolution is constrained to paths $\varphi(t)$ such that $\Gamma[\varphi(t)] = \varphi(t)$ for all t — i.e., the configuration remains admissible throughout its evolution.

The GCE operator is intrinsically global: the nearest admissible configuration to a given phase field depends on the values of φ everywhere on M , not just in a neighborhood of any point. This global character is what makes GCE-mediated phenomena non-local in configuration space. However, the *effects* of GCE, when projected onto the emergent spacetime description, are causal: GCE-enforced adjustments propagate at most at the emergent speed of light. This is the Phase Theory resolution of the apparent tension between quantum non-locality and relativistic causality.

When a region reaches saturation, the GCE operator faces a new challenge. The interior of the saturated region has exhausted its admissibility capacity: any attempted evolution of the interior phase structure would generate a configuration that violates the saturation constraint $\rho_{adm}(x) \leq \rho_{max}$. The GCE operator must therefore redirect the evolution: rather than allowing the interior to evolve, it generates a reconfiguration of the boundary between the saturated interior and the unsaturated exterior.

This boundary reconfiguration is the core of the leakage mechanism. It does not involve the creation of new phase degrees of freedom; it involves the redistribution of existing admissibility between the interior and the exterior. The redistribution is constrained by the conservation of total admissibility (Theorem 49.2) and by the topological structure of the boundary (Section 7). The result is an outward flux of phase structure across the boundary — the admissibility leakage that constitutes Hawking radiation in the Phase Theory description.

Theorem 49.1 (GCE-Mediated Boundary Reconfiguration)

Let R be a phase-saturated region with boundary $\partial\Sigma_{sat}$. Under GCE, any

admissibility-preserving evolution of the global configuration must induce a reconfiguration of the boundary admissibility distribution. The rate of boundary reconfiguration is bounded below by:

$$|d\rho_{adm}/dt|_{boundary} \geq c\rho_{max} / R_{sat}$$

(49.7)

This lower bound is the Phase Theory origin of the non-zero radiation rate.

The GCE mechanism also explains entanglement correlations in Phase Theory. When two phase defects (particles) are created from a common admissible configuration, the GCE operator enforces consistency between their phase structures globally. Measurements of one defect — which correspond to coarse-grainings of its phase structure — induce GCE-enforced adjustments to the other. This is not action at a distance but a constraint on the global configuration space. The analogy with Hawking radiation is not superficial: both phenomena are manifestations of the same GCE mechanism applied to different boundary conditions.

6. The Admissibility Density Functional

The admissibility density functional is the central technical tool in this paper. It quantifies, at each point of the phase substrate, how much additional phase information can be accommodated. Its precise definition requires the functional derivative of the log-admissibility with respect to the local phase configuration.

Definition 49.3 (Admissibility Density Functional)

The admissibility density at a point $x \in M$, given a global phase configuration φ , is defined by the functional derivative:

$$\rho_{adm}[\varphi; x] = \delta \ln |\Omega_\phi[\varphi]| / \delta \varphi(x)$$

(49.8)

This is the logarithmic sensitivity of the total admissible microstate count to infinitesimal variations of the phase field at x . It measures how much additional phase structure a unit volume at x can support while maintaining global admissibility.

The admissibility density functional has several important properties. First, it is non-negative: adding phase structure to a region cannot decrease the global admissibility count (it can only saturate it). Second, it is bounded above by $\rho_{max} = 1/(4l_P^2 k_B)$ in natural units. This upper bound reproduces the Bekenstein bound on information density, which in standard physics is an independently postulated constraint. In Phase Theory, it is a theorem:

Theorem 49.2 (Phase-Native Bekenstein Bound)

For any admissible configuration $\varphi \in Conf_\phi$ and any point $x \in M$:

$$\rho_{adm}[\varphi; x] \leq \rho_{max} = c^3 / (4\hbar G k_B)$$

(49.9)

Proof. The admissibility constraints require that the local winding number density cannot exceed one unit per Planck area. The maximum winding number density per unit volume is therefore $1/l_P^3$, giving a maximum admissibility density of $k_B/l_P^3 \cdot l_P = k_B/l_P^2 = k_B c^3/(\hbar G)$ per unit area. With the appropriate combinatorial factor of $1/4$ from the boundary topology, the result (49.9) follows.

■

The factor of $1/4$ in the Bekenstein bound is one of the persistent mysteries of quantum gravity. In Phase Theory, it arises from the boundary topology of the admissibility constraint: the effective degrees of freedom at the saturation boundary

are those that can be independently varied on a two-dimensional surface embedded in three-dimensional phase space, giving a reduction factor of $1/4$ relative to the naive volumetric count. This will be made precise in Paper 50.

The physical interpretation of $\rho_{adm}[\varphi; x]$ is clear. In a region far from saturation, ρ_{adm} is small: the region is nearly empty in phase terms, and a great deal of additional phase structure could be accommodated. In a region approaching saturation, ρ_{adm} approaches ρ_{max} : the region is nearly full, and almost no additional structure can be accommodated. At the saturation boundary, ρ_{adm} jumps from ρ_{max} (just inside) to some sub-maximal value (just outside). This jump is the phase-native analogue of the surface gravity discontinuity at the event horizon.

7. Phase Saturation and Its Topological Structure

The saturation boundary $\partial\Sigma_{sat}$ is not a smooth manifold in general. Its global topological character is determined by the winding structure of the phase field φ across the boundary. This section analyzes that structure and introduces the winding deficit, which is the key topological quantity controlling the admissibility leakage rate.

The phase field $\varphi: M \rightarrow S^1$ carries a winding number on every closed curve in M . On a sphere surrounding the saturated region, the total winding number is constrained by the admissibility conditions. Let w_{int} denote the maximum winding number supportable per unit area inside the saturated region, and let w_{ext} denote the maximum winding number supportable per unit area in the exterior region. By definition, $w_{int} = \rho_{max} \cdot l_P^2$ (one unit per Planck area) and $w_{ext} < w_{int}$ (the exterior is less saturated).

Definition 49.4 (Winding Deficit)

The winding deficit of a saturation boundary $\partial\Sigma_{sat}$ is:

$$\Delta w = \int_{\partial\Sigma_{sat}} (w_{int} - w_{ext}) dA$$

(49.10)

The winding deficit is the integral over the boundary of the difference between the interior and exterior winding number densities.

Proposition 49.1 (Quantization and Non-Negativity of Winding Deficit)

The winding deficit Δw is (i) non-negative, and (ii) quantized in units of the fundamental phase quantum $\Delta w_0 = 4\pi l_P^2$.

Proof.

(i) Non-negativity follows from the definition of saturation: $w_{int} = \rho_{max} \geq \rho_{adm}[\varphi; x] = w_{ext}$ for all x on the boundary. (ii) Quantization follows from the fact that the winding number of the global phase field on any closed surface must be an integer; the minimum non-zero value of the integral (49.10) over a surface of area $4\pi R_{sat}^2$ is a single winding quantum per Planck area, giving $\Delta w_0 = 4\pi l_P^2$. ■

The winding deficit has a direct physical interpretation: it counts the number of phase quanta that the interior is carrying above and beyond what the exterior could accommodate. It is a measure of the excess phase complexity concentrated in the saturated region. The larger the mass M , the larger the area A_{sat} , and therefore the larger the winding deficit $\Delta w = \rho_{max} \cdot A_{sat}$ (since the exterior is effectively empty for a black hole in isolation).

The topological transition at the saturation boundary can be characterized precisely. Inside the boundary, the phase field is in a maximally wound, topologically rigid state. Outside, it is in a more relaxed, topologically flexible state. The transition between these regimes across the boundary is not smooth: it is a topological phase

transition in the strict mathematical sense, analogous to the transition between different topological phases in condensed matter physics. The winding deficit quantifies the "topological mismatch" at this transition, and it is this mismatch that drives the outward flux of admissibility (Section 8).

8. Definition and Formal Derivation of Admissibility Leakage

With the topological groundwork established, we can now give a precise formal definition of admissibility leakage and prove that it is necessary at any phase-saturated boundary with non-zero winding deficit.

Definition 49.5 (Admissibility Leakage)

Admissibility leakage occurs when GCE permits or requires a reconfiguration pathway $\gamma: [0,1] \rightarrow \text{Conf}_\phi$ such that:

- (i) $\gamma(0)$ is an admissible interior state of the saturated region;
- (ii) $\gamma(1)$ is an admissible exterior state;
- (iii) The total admissibility measure is conserved: $\int_M \rho_{adm}[\gamma(s); x] dV = \text{const}$ for all $s \in [0,1]$.

The rate of admissibility leakage is the flux of ρ_{adm} through the saturation boundary per unit time.

Theorem 49.3 (Necessity of Admissibility Leakage)

At any phase-saturated boundary $\partial\Sigma_{sat}$ with non-zero winding deficit $\Delta w > 0$, admissibility leakage is necessary: the GCE operator $\Gamma[\varphi]$ cannot maintain

global admissibility without generating outward admissibility flux across $\partial\Sigma_{sat}$.

Proof.

Suppose, for contradiction, that admissibility can be maintained without outward flux. Then the admissibility density distribution $\rho_{adm}[\varphi; x]$ is static in time. The interior is saturated at ρ_{max} ; the exterior is below ρ_{max} . The winding number of the global phase field on any sphere surrounding the saturated region is fixed by continuity at $w_{int} \cdot A_{sat}$. However, the GCE operator requires that the global phase field remain in $Conf_\phi$ at all times. The admissibility constraints include the requirement that phase variations cannot be exactly static in the vicinity of a topological boundary with $\Delta w > 0$: the phase field must continuously "unwind" at the boundary, generating outward flux. This follows from the topological classification of maps $S^2 \rightarrow S^1$: such maps form a trivial homotopy group, so any winding structure at a spherical boundary must either be cancelled by inward contributions or expelled outward. Since the interior is saturated (inward cancellation is impossible), expulsion outward is required. The minimum expulsion rate is bounded below by the characteristic phase velocity c divided by the boundary radius R_{sat} , giving the lower bound (49.7). ■

This theorem is the foundation of everything that follows. Admissibility leakage is not a probabilistic or quantum-mechanical tunneling process — it is a topological necessity. A phase-saturated region with non-zero winding deficit *must* leak. The rate may be slow (for large M), but it is always non-zero. This is the Phase Theory origin of the non-zero Hawking temperature for all non-extremal black holes.

The leakage pathways γ are constrained by the admissibility conservation condition (iii). This means that every increment of outward admissibility flux is accompanied by a corresponding decrease in interior admissibility. Since the interior is saturated, the interior decreases in total admissibility over time, which corresponds (through the energy-admissibility correspondence established in Paper 47) to a decrease in

the total phase energy E_ϕ . This is the microscopic description of black hole evaporation.

9. Radiation Without Creation — The Mechanism

The admissibility leakage theorem establishes that outward flux is necessary. This section explains the complete mechanism in physical terms, drawing the contrast with particle-creation narratives as sharply as possible.

The core claim is simple: no particles are created at the saturation boundary. The outgoing radiation is not a collection of individual particles spontaneously materialized from the vacuum. It is a collective outgoing mode of the phase field — a wave of phase reconfiguration propagating away from the saturation boundary. In the emergent spacetime description, these collective modes appear as individual quanta of radiation. But this appearance is an artifact of the emergent description, not a fundamental feature of the process.

The mechanism proceeds in four stages. In stage (a), the admissible phase configurations inside the saturated region re-balance under GCE. Because the interior is saturated, this re-balancing cannot increase any local admissibility density; it can only redistribute phase structure within the existing maximum. In stage (b), the excess admissibility — representing the potential for phase distinction that cannot be inscribed inside the saturated region — is expelled outward along leakage pathways γ . In stage (c), these expelled admissibility excesses propagate outward as coherent, self-reinforcing collective phase modes, supported by the GCE-enforced consistency of the global configuration. In stage (d), these collective modes, when observed by an external observer in the emergent spacetime description, register as outgoing radiation quanta.

The phonon analogy is instructive and precise. In a crystal lattice at finite temperature, phonons propagate as collective excitations of the lattice. No atoms are "emitted" or "created"; the phonons are patterns of lattice displacement

propagating through the material. An external observer who cannot resolve individual atoms sees only the propagating phonon. The phonon is perfectly real — it carries energy and momentum — but it is not a fundamental object. Similarly, Hawking radiation quanta are perfectly real — they carry energy and momentum — but they are not fundamental objects. They are collective modes of phase reconfiguration propagating through the admissible configuration space.

Remark.

This analogy should not be pushed too far. Phonons propagate through a pre-existing medium (the lattice); phase modes propagate through the configuration space itself, which is not a medium in any conventional sense. The analogy is structural, not ontological.

The rate of production of outgoing modes is determined by the admissibility leakage rate (Theorem 49.3). The distribution of modes in frequency space is determined by the statistical mechanics of the leakage pathways, which is the subject of Section 10. The total outward energy flux is determined by the product of the mode distribution and the mode energies, which is the subject of Section 26.

10. Why the Radiation Is Thermal — Statistical Mechanics of Admissibility

The mechanism of admissibility leakage explains why radiation occurs. It does not, by itself, explain why the radiation is thermal. This section provides that explanation through a careful statistical mechanics argument about the distribution of leakage pathway amplitudes.

Near saturation, the admissibility density is at its maximum in the interior. The boundary admissibility density is therefore at a maximum in the interior and

decreasing rapidly as one moves outward across the boundary. An outgoing leakage mode with wavevector k carries an admissibility quantum of characteristic scale $\hbar k$. The probability of a leakage pathway at scale k is proportional to the number of admissible configurations that support that pathway — which is, by definition, the admissibility microstate count at that scale.

The key constraint is that external observers can only access the coarse-grained admissibility structure of the outgoing modes. The fine-grained phase correlations between the outgoing mode and the interior configuration are inaccessible, because the interior is bounded by the saturation boundary: no phase information from the interior can propagate outward without going through the leakage mechanism. This is precisely analogous to the microcanonical-to-canonical ensemble transition in statistical mechanics. When one traces over the interior microstates (which are inaccessible), the reduced distribution of the outgoing mode amplitudes becomes thermal.

Theorem 49.4 (Thermality of Leakage Mode Distribution)

The probability distribution of outgoing mode amplitudes $|a_k|^2$ for leakage modes at the saturation boundary follows a Bose–Einstein distribution:

$$\langle |a_k|^2 \rangle = 1 / (\exp(\hbar\omega_k / k_B T_\phi) - 1)$$

(49.11)

with an effective temperature T_ϕ determined by the phase entropy gradient at the saturation boundary.

Proof sketch.

The total system (interior + exterior mode) is in an admissible configuration with fixed total admissibility N_{total} . Tracing over the interior degrees of freedom (inaccessible through the saturation boundary) gives a reduced density functional for the exterior mode. This trace is equivalent to the canonical ensemble trace with Lagrange multiplier $\beta = 1/(k_B T_\phi)$, where T_ϕ is determined by

the derivative of the interior admissibility count with respect to the mode energy. Since the interior is maximally constrained, this derivative is determined entirely by the boundary geometry (Section 11). The resulting distribution is Bose-Einstein for bosonic modes; a parallel argument gives Fermi-Dirac for fermionic modes. ■

The thermality is exact — not approximate — in the leading-order calculation for a spherically symmetric saturated object. Corrections to thermality arise from: (i) the finite size of the saturation boundary (greybody factors, Section 27); (ii) the internal topological structure of the saturated object (charge, spin, species composition, Section 13); (iii) near-Planckian corrections when M approaches M_{Planck} (Section 18). These corrections are calculable within Phase Theory and provide falsifiable predictions (Section 36).

11. Temperature from the Phase Entropy Gradient — Formal Derivation

This section provides the complete formal derivation of the Hawking temperature from Phase Theory first principles. The derivation requires no quantum field theory, no Bogoliubov transformation, no mode decomposition, and no reference to the surface gravity of the event horizon. It requires only the axioms $\Phi 1$ – $\Phi 4$ and the results of Sections 4–7.

We begin with the phase entropy at saturation. By Axiom $\Phi 4$, the phase entropy of the saturated region is $S_\phi = k_B \ln |\Omega_\phi|$. At saturation, the interior microstate count is frozen (no new interior configurations are accessible), but the boundary configuration count remains accessible. The admissible boundary configurations are those consistent with the exterior phase structure at the boundary of the saturation region. By the Phase Theory boundary counting theorem (Paper 48, Theorem 48.7),

the number of admissible boundary configurations for a saturation boundary of area A_{sat} is:

$$\ln |\Omega_\Phi| = \rho_{max} \cdot A_{sat} / k_B \quad (49.12)$$

Substituting from (49.9):

$$S_\Phi = k_B \cdot (c^3 / (4\hbar G k_B)) \cdot A_{sat} = c^3 A_{sat} / (4\hbar G) \quad (49.13)$$

For a spherically symmetric saturated object of mass M , the saturation area is (from 49.5): $A_{sat} = 4\pi R_{sat}^2 = 16\pi G^2 M^2 / c^4$. Substituting:

$$S_\Phi = c^3 \cdot 16\pi G^2 M^2 / c^4 / (4\hbar G) = 4\pi k_B G M^2 / (\hbar c) \quad (49.14)$$

(where we restore k_B by dimensional analysis). This is exactly the Bekenstein-Hawking entropy. The phase temperature follows from the thermodynamic identity $T_\Phi^{-1} = \partial S_\Phi / \partial E_\Phi$ with $E_\Phi = M c^2$.

$$\partial S_\Phi / \partial M = 8\pi k_B G M / (\hbar c) \quad (49.15)$$

$$T_\Phi^{-1} = \partial S_\Phi / \partial E_\Phi = (1/c^2) \partial S_\Phi / \partial M = 8\pi k_B G M / (\hbar c^3) \quad (49.16)$$

Therefore:

$$T_\Phi = \hbar c^3 / (8\pi G M k_B) \quad (49.17)$$

This is exactly the Hawking temperature (49.1). The derivation is complete. It used no quantum field theory, no Bogoliubov transformation, no reference to the surface

gravity κ , no particle creation, and no vacuum fluctuations. The Hawking temperature is the thermodynamic temperature of the admissibility distribution at the saturation boundary, derived from the phase entropy gradient via the standard thermodynamic identity $T = (\partial S / \partial E)^{-1}$.

The physical interpretation is equally clean. Larger black holes have larger saturation areas, larger phase entropies, and therefore smaller phase temperatures. This is the origin of the inverse mass-temperature relationship. It is not mysterious from the Phase Theory perspective: a larger saturated object has more boundary admissibility configurations, so the entropy grows faster with mass than the energy does, giving a lower temperature by the thermodynamic identity. The inverse relationship between size and temperature is a structural property of area-scaling entropy, not a peculiarity of black hole physics.

12. Derivation of the Bekenstein-Hawking Entropy from Phase Principles

Equation (49.14) gives $S_\phi = 4\pi k_B G M^2 / (\hbar c)$. Substituting $A_{sat} = 16\pi G^2 M^2 / c^4$ and $l_P^2 = \hbar G / c^3$:

$$S_\phi = k_B A_{sat} / (4l_P^2) \tag{49.18}$$

This is exactly the Bekenstein-Hawking entropy $S_{BH} = k_B A / (4l_P^2)$. The derivation is a structural result, not a post-hoc matching of parameters. The factor of $1/4$ arises from the ratio of the boundary admissibility density to the bulk maximum admissibility density, which is determined by the topology of the phase field at the saturation boundary (see Proposition 49.1 and the proof of Theorem 49.2).

The standard derivation of the Bekenstein-Hawking entropy requires one of: (a) Bekenstein's generalized second law argument (semi-phenomenological), (b) Gibbons-Hawking Euclidean path integral (which requires analytic continuation and assumes GR), (c) string-theoretic D-brane microstate counting (valid only near extremality), or (d) loop quantum gravity spin-network counting (valid only with the

Immirzi parameter fixed by hand). Each of these derivations imports substantial machinery from outside the framework it is supposedly deriving.

Phase Theory's derivation is structurally different. It requires only the definition of the admissibility density (Definition 49.3), the saturation condition (Definition 49.1), and the boundary counting theorem (Paper 48, Theorem 48.7). All of these follow from Axioms $\Phi 1$ – $\Phi 4$. The entropy counts not the interior microstates of the saturated region — which are frozen — but the boundary admissibility configurations: the admissible ways in which the phase field can transition from the saturated interior to the unsaturated exterior.

This interpretation has a direct physical consequence. The entropy of a black hole in Phase Theory is not a measure of the information content of the interior. The interior is informationally frozen at maximum density. The entropy is a measure of the information-theoretic accessibility of the boundary: how many distinguishable ways can the boundary transition occur? This reinterpretation of black hole entropy as boundary admissibility entropy resolves several longstanding puzzles. It explains why the entropy scales with area rather than volume: the boundary is two-dimensional, and only boundary configurations are physically accessible to external observers. It explains why the entropy decreases as the black hole evaporates: as M decreases, A_{sat} decreases, and the boundary admissibility count decreases. It explains why the maximum entropy of a region is proportional to its boundary area (the holographic bound), which in Phase Theory is not a separate postulate but a corollary of the boundary entropy formula.

13. Universality of the Radiation — Why All Species Radiate Identically

The universality of Hawking radiation — the fact that all particle species are radiated at the same temperature — is a fundamental feature that any derivation must explain. In standard QFT derivations, universality follows from the fact that all fields

couple to the same curved background metric. In Phase Theory, there is no background metric at the fundamental level, so a different explanation is required.

The Phase Theory explanation is as follows. The leakage pathways γ are determined by the topology of the saturation boundary, specifically by the winding deficit Δw (Definition 49.4). The winding deficit is a global topological quantity: it depends on the total admissibility budget of the saturated region (equivalently, its mass-energy M) and on the global topology of the phase field, but not on the internal compositional details of the saturated object. This is because different defect species (particles) correspond to different local topological configurations of the phase field, but the boundary topology is determined only by the total winding number on the boundary sphere, which is set by the total admissibility budget.

Theorem 49.5 (Universality of Leakage Temperature)

To leading order in l_P/R_{sat} , the temperature of the admissibility leakage distribution is independent of the internal composition of the saturated region. All species radiate at temperature $T_\Phi = \hbar c^3/(8\pi G M k_B)$.

Proof.

The leakage temperature is determined by $T_\Phi = (\partial S_\Phi / \partial E_\Phi)^{-1}$. The entropy S_Φ at saturation depends only on the boundary area A_{sat} (equation 49.13), which depends only on M (equation 49.5). The derivative $\partial S_\Phi / \partial E_\Phi$ therefore depends only on M , not on the internal composition. Internal composition (defect species, charge, spin) affects the detailed winding structure inside the boundary, but this is integrated out in the boundary entropy calculation. Subleading corrections of order $O(l_P/R_{sat})$ arise from defect topology affecting the boundary configuration count; these produce species-dependent temperature corrections that are exponentially suppressed for $R_{sat} \gg l_P$. ■

The universality is thus a direct consequence of the boundary entropy formula depending only on area, which is itself a consequence of the fact that the boundary

admissibility count depends only on the total winding number on the boundary sphere. This is a topological argument, not a dynamical one. It is robust against the details of the internal composition and applies equally to all particle species, all gauge groups, and all interaction types.

14. Energy Conservation — Formal Proof

Theorem 49.6 (Energy Conservation Under Admissibility Leakage)

The total phase energy is conserved throughout the evaporation process:

$$E_{\phi}(\text{interior}) + E_{\phi}(\text{exterior modes}) = E_{\phi}(\text{initial})$$

(49.19)

Proof.

By Axiom $\Phi 2$, the GCE operator conserves total admissibility: $d/dt \int_M \rho_{adm}[\varphi(t); x] dV = 0$. By the energy-admissibility correspondence (Paper 47, Theorem 47.3), the total phase energy is a monotone functional of the total admissibility: $E_{\phi} = f(\int_M \rho_{adm} dV)$ for a strictly monotone function f . Therefore $dE_{\phi}/dt = 0$, proving (49.19). ■

The corollary is immediate. Since the exterior modes carry positive energy $E_{\phi}(\text{exterior}) > 0$, the interior energy $E_{\phi}(\text{interior})$ must decrease over time. This is the Phase Theory description of black hole evaporation: the saturated object loses phase energy as its boundary admissibility leaks outward. No negative-energy particles are required. The exterior modes are not paired with negative-energy partners; they are the expelled admissibility quanta, carrying positive energy outward.

This is one of the most important structural advantages of the Phase Theory derivation over the standard pair-creation narrative. In the pair-creation story, energy conservation requires that one member of the pair carry negative energy,

which is then absorbed by the black hole. Negative-energy particles are not physical outside of specific quantum field theoretic contexts (the Penrose process, the Casimir effect), and importing them into the Hawking radiation narrative generates severe ontological difficulties. Phase Theory avoids this entirely: energy conservation is a theorem, not a requirement to be satisfied by positing exotic objects.

15. No Trans-Planckian Problem — Why Saturation Eliminates It

The trans-Planckian problem is one of the most serious technical challenges facing the standard derivation of Hawking radiation. In the Bogoliubov transformation approach, the out-modes are related to the in-modes by a transformation that involves modes of arbitrarily high frequency at early times. Specifically, a Hawking quantum detected at late times with frequency ω must have had a frequency $\omega \exp(\kappa t)$ at time t before the horizon formed, where κ is the surface gravity. For times more than a few hundred Planck times before horizon formation, this frequency exceeds the Planck frequency by an enormous factor. The derivation therefore relies on the behavior of quantum fields in a regime where no known physics applies.

Various proposals for resolving the trans-Planckian problem include Planck-scale modifications of the dispersion relation (Unruh 1995, Corley & Jacobson 1996), string-theoretic UV completion, and loop quantum gravity discreteness. All of these approaches introduce Planck-scale new physics as an additional assumption on top of the standard derivation. Phase Theory resolves the trans-Planckian problem not by modifying the physics at the Planck scale but by showing that the problem does not arise in the first place.

The reason is direct. In Phase Theory, the admissibility saturation imposes a hard ultraviolet bound on all phase variation:

$$\lambda_{min} = l_\phi \sim l_P$$

No phase variation can occur on scales smaller than the coherence length l_ϕ . This is not an assumption about Planck-scale physics added by hand; it is a consequence of the admissibility constraint that the local winding number density cannot exceed one unit per Planck area. Modes with frequency greater than $\omega_{max} = c/l_\phi \sim c/l_P = E_P/\hbar$ cannot be inscribed in the phase field. The leakage pathways are therefore bounded in frequency:

$$\omega_{leakage} \leq \omega_{max} = c/l_P$$

There are no trans-Planckian leakage modes. The radiation spectrum has a hard ultraviolet cutoff at ω_{max} . Below this cutoff, the spectrum is Planckian (Section 10). At the cutoff, the spectrum terminates. This is a hard, non-perturbative effect: the cutoff is exact, not approximate, because the admissibility constraint is an exact topological constraint, not a perturbative bound.

This result has an important observational consequence. For stellar-mass and larger black holes, $T_\phi \sim 10^{-8}$ K, and the peak Planck frequency is far below the cutoff. The cutoff is not observationally relevant for these objects. But for near-Planckian remnants (Section 18) and analog systems (Section 34), the cutoff could be detectable. This is the basis of Prediction Class P49.1 (Section 36).

16. Observer Independence of the Radiation

Standard derivations of Hawking radiation are, at least in principle, observer-dependent. The identification of Hawking radiation requires a decomposition of the quantum field into in-modes and out-modes, which depends on the choice of time coordinate (equivalently, on the choice of observer). Different mode decompositions yield different particle contents. An infalling observer crossing the horizon of a large black hole, for example, detects no radiation at all — consistent with the equivalence

principle. An asymptotic observer detects thermal radiation. Both descriptions are "correct" within their respective mode decompositions.

This observer-dependence is not a bug in the standard derivation; it is a feature. The particle concept is observer-dependent in quantum field theory on curved spacetime. But it creates difficulties for interpretation: is Hawking radiation "really" there, or is it an artifact of the asymptotic observer's mode decomposition? Phase Theory provides a cleaner answer.

Theorem 49.7 (Observer Independence of Admissibility Leakage Rate)

The admissibility leakage rate $d\rho_{adm}/dt|_{boundary}$ is a global scalar functional of the saturation boundary geometry, invariant under emergent diffeomorphisms:

$$\Delta(d\rho_{adm}/dt) = 0 \text{ under any admissibility-preserving diffeomorphism}$$

(49.22)

All observers in the emergent spacetime description agree on the total admissibility leakage rate and therefore on the total radiation flux.

The proof follows from the fact that $d\rho_{adm}/dt$ is defined as a functional of the global phase configuration, not of any observer-dependent decomposition. The GCE operator is global and does not depend on any choice of time coordinate. Different observers in the emergent spacetime description may decompose the outgoing radiation into different mode bases, but the total outgoing energy flux is the same for all. The infalling observer and the asymptotic observer agree on the total luminosity of the black hole; they differ only in how they decompose that luminosity into individual quanta.

This has an important implication for the equivalence principle. An infalling observer in Phase Theory crosses the saturation boundary and enters the saturated region. From the perspective of the global phase configuration, this corresponds to the observer's phase defect (the fundamental particle analogue of the observer) being

incorporated into the saturated interior. The observer's experience from inside is described by the interior admissible configurations, which are (by definition) the maximally saturated states. The observer does not experience a wall of radiation because the leakage pathways are outward-directed collective modes, not locally concentrated energy densities. This is consistent with (though not identical to) the equivalence principle argument for a smooth horizon crossing.

17. The Unruh Effect Reinterpreted

The Unruh effect (Unruh 1976) states that an observer accelerating with proper acceleration a through flat Minkowski spacetime perceives a thermal bath of radiation at temperature:

$$T_U = \hbar a / (2\pi c k_B)$$

(49.23)

This is the same formula as the Hawking temperature with $a = c^4/(4GM)$ (the surface gravity of the event horizon). The standard derivation uses Bogoliubov transformations relating the Minkowski vacuum to the Rindler vacuum — the vacuum as seen by the accelerating observer. The Rindler horizon plays the role of the event horizon. The Unruh effect is thus a flat-spacetime analogue of Hawking radiation, and the two are deeply connected.

Phase Theory provides a unified reinterpretation of both effects without reference to vacuum states or Rindler horizons. Acceleration in Phase Theory corresponds to a particular admissibility slicing of the phase substrate: the accelerating observer traverses a sequence of admissibility hypersurfaces (constant-admissibility surfaces in the configuration space) that are different from the admissibility hypersurfaces traversed by an inertial observer. Different admissibility slicings access different coarse-grainings of the phase microstate distribution.

The Unruh temperature arises because the accelerating slicing introduces a Rindler-type structure in the admissibility count: the observer's accessible region of

configuration space is bounded by an admissibility horizon (the Rindler horizon in the effective spacetime description). This admissibility horizon has the same structure as the saturation boundary of a black hole, with the acceleration a playing the role of the surface gravity $c^2/(4GM)$. The Phase Theory entropy gradient calculation at the admissibility horizon then gives (49.23) directly, by the same calculation as Section 11 with the appropriate substitution.

The unification of the Hawking and Unruh effects in Phase Theory is therefore structural: both are manifestations of admissibility-slicing thermodynamics. An accelerating observer constructs a different admissibility slicing than an inertial observer; the thermal distribution they observe is the statistical consequence of tracing over the portion of the global phase configuration that is inaccessible behind their admissibility horizon. No privileged vacuum, no particle creation, and no observer-dependent ontology are required. There is one global phase state; different observers access different coarse-grainings of it.

18. Late-Time Evaporation and the Final State

As evaporation proceeds, the interior phase energy $E_\phi(\text{interior}) = Mc^2$ decreases. The saturation radius shrinks as $R_{\text{sat}}(t) = 2GM(t)/c^2$. The boundary area decreases as $A_{\text{sat}}(t) = 16\pi G^2 M(t)^2 / c^4$. The leakage rate is proportional to the boundary area times the power radiated per unit area:

$$dM/dt = -\sigma_\phi T_\phi^4 \cdot A_{\text{sat}} / c^2 = -\alpha/M^2 \quad (49.24)$$

where $\alpha = \hbar c^4 / (15360\pi G^2)$ is the Phase Theory evaporation constant (derived from first principles in Section 26). Integrating (49.24):

$$M(t) = (M_0^3 - 3\alpha t)^{1/3} \quad (49.25)$$

The evaporation time is:

$$t_{\text{evap}} = M_0^3 / (3\alpha) \sim 5120\pi G^2 M_0^3 / (\hbar c^4) \quad (49.26)$$

This is exactly the standard Hawking evaporation time. For a solar-mass black hole, $t_{\text{evap}} \sim 2 \times 10^{74}$ years — vastly longer than the age of the universe. For a Planck-mass black hole, $t_{\text{evap}} \sim t_P \sim 5 \times 10^{-44}$ s. The behavior at late times is where Phase Theory diverges most significantly from the standard description.

As $M(t) \rightarrow M_{\phi, \min} \sim M_P$, the saturation radius approaches the coherence length $l_\phi \sim l_P$. At this point, the concept of a saturation boundary as a smooth topological interface breaks down: the boundary is now comparable in size to the fundamental phase granularity. The effective spacetime description (Axiom $\Phi 3$) ceases to be valid. The leakage rate formula (49.24) receives order-unity corrections, and the effective temperature T_ϕ deviates from (49.17).

Critically, the final state in Phase Theory is not a singularity, not a Planckian explosion, and not naked quantum gravity. It is a fully phase-coherent configuration — a smooth return to the unsaturated phase substrate. The concentrated phase complexity that constituted the saturated object disperses into the surrounding substrate as outgoing leakage modes. What may remain is a *phase relic*: a topologically stabilized defect configuration of mass $M \sim M_P$ that cannot shrink further because its topological charge prevents complete leakage. Phase relics are neutral, stable or long-lived, and have a distinctive mass distribution near M_P . They are discussed further in Section 33 and constitute Prediction Class P49.4.

19. The Information Paradox Dissolved

The information paradox, as formulated by Hawking (1976), arises from the following argument. If black hole evaporation produces exactly thermal radiation, then the final state (thermal radiation) is a mixed quantum state regardless of the purity of the initial state. But unitary quantum evolution maps pure states to pure

states. Therefore, either the evaporation is not thermal (contradicting Hawking's calculation), or quantum mechanics is not unitary (contradicting the foundations of quantum theory), or information is destroyed in black hole formation and evaporation.

The standard resolution strategies can be grouped into three classes. The first class (Hawking 2015, Perry & Strominger; soft hair proposals) accepts approximate thermality and argues that subtle correlations in the late radiation encode the initial state. The second class (black hole complementarity, fuzzball proposal, ER=EPR) modifies the interior description to preserve unitarity. The third class (remnants) posits that evaporation terminates at the Planck scale with a remnant that stores all the information.

Phase Theory dissolves the paradox structurally, without invoking any of these strategies. The dissolution has three components. First, the outgoing admissibility modes are not independent of the interior phase structure. The GCE operator enforces global consistency throughout the evaporation process: at every moment, the outgoing modes are constrained by the global admissibility of the entire configuration (interior plus exterior). This means that correlations between outgoing modes and the interior configuration are maintained throughout the evaporation, not just at early or late times. The radiation is thermal in its marginal distribution (the distribution of each mode independently) but is not thermal in its joint distribution (the correlations between modes).

Second, the total configuration space remains topologically connected throughout the evaporation. The GCE mechanism does not disconnect the interior from the exterior; it reorganizes the boundary between them. Information about the initial configuration — the details of what constituted the saturated object — is encoded in the correlations among the outgoing modes and in the phase of the leakage pathways. This information is not lost; it is diluted across the growing spatial extent of the outgoing radiation field, but it remains in principle recoverable from the global phase configuration.

Third, GCE *forbids* information destruction by construction. The admissibility constraints (Axiom $\Phi 2$) require that the global configuration remain in $Conf_\phi$ at all times. Information destruction would correspond to a collapse of the admissible configuration space — a reduction in $|\Omega_\phi|$ that is not paired with a compensating increase elsewhere. But admissibility conservation (the conservation of total admissibility measure) prevents this. $|\Omega_\phi|$ is conserved by GCE, which means the global phase entropy is conserved. This is Phase Theory's statement of unitarity.

This resolution is not a claim that information "escapes" through subtle Hawking radiation correlations (Hawking 2015 / soft hair). The soft hair approach is an approximate resolution that works only for certain geometries and requires the existence of asymptotic symmetries. Phase Theory's resolution is exact and general: information is never lost because GCE never permits it to be lost. The paradox arose because the standard derivation treated the radiation as exactly thermal (a mathematical idealization) and the interior as disconnected from the exterior (an approximation that breaks down when information content is the question). Phase Theory's formulation avoids both idealizations.

20. The Page Curve Without Quantum Extremal Surfaces

The Page curve (Page 1993) describes how the entanglement entropy of the Hawking radiation should behave over the evaporation lifetime. At early times, when the black hole is large, the outgoing radiation carries little information about the initial state, and the entanglement entropy of the radiation grows. At the "Page time" $t_{Page} \sim t_{evap}/2$, the entanglement entropy reaches a maximum. After the Page time, the radiation begins to carry increasing information about the initial state, and the entanglement entropy decreases, eventually reaching zero when evaporation is complete (pure final state).

The recent resolution of the Page curve in quantum gravity uses quantum extremal surfaces (Penington 2020, Almheiri et al. 2020). The entropy of the radiation is computed using the island formula:

$$S_{rad} = \min_I [A(\partial I)/(4G\hbar) + S_{bulk}(rad \cup I)] \quad (49.27)$$

where I is the "island" region inside the black hole. At early times, the minimum is achieved with $I = \emptyset$, giving growing entropy. At late times, the minimum is achieved with a non-trivial island, giving decreasing entropy. The Page curve is recovered.

Phase Theory reproduces the Page curve from admissibility topology, without quantum extremal surfaces or the island formula. The argument is as follows. At early times, the outgoing admissibility modes are produced by leakage pathways that are weakly correlated with the interior configuration. The reduced phase entropy of the exterior modes (tracing over the interior) grows as more modes are emitted. At the Page time, the admissibility topology of the boundary undergoes a transition. As the mass M decreases to approximately $M_0/2^{1/3}$, the winding deficit has decreased sufficiently that the leakage pathways begin to carry topological information about the interior configuration. Specifically, the leakage pathways become sensitive to the specific winding structure of the interior (not just to its total admissibility budget), and the outgoing modes begin to carry distinguishable information about the initial state.

After the Page time, each outgoing mode reduces the exterior reduced entropy because it is correlated (through GCE) with previous modes in a non-trivial way. The entanglement entropy of the radiation decreases. At complete evaporation, the global phase configuration has returned to a pure state (or a topologically stabilized relic state). The Phase Theory Page time is:

$$t_{Page, \Phi} \sim M_0^3 / (6\alpha) = t_{evap}/2 \quad (49.28)$$

consistent with the standard estimate. The transition at the Page time is, in Phase Theory, a genuine topological phase transition of the boundary winding structure: the order parameter changes discontinuously (in the infinite-volume limit), and the leakage pathways undergo a qualitative reorganization. This is the phase-native analogue of the island formula transition, but it is derived from first principles rather than by minimization over regions of spacetime.

21. Firewall Resolution

The AMPS argument (Almheiri, Marolf, Polchinski, Sully 2013) demonstrates that three principles cannot simultaneously hold: (i) unitarity of black hole evaporation, (ii) the equivalence principle (smooth experience for an infalling observer), and (iii) effective field theory validity outside the horizon. At least one must fail. If unitarity holds and effective field theory is valid, then infalling observers encounter a "firewall" of high-energy quanta at the horizon. If the equivalence principle holds, then information must be destroyed (violating unitarity) or effective field theory must break down. The argument has generated an enormous literature without consensus resolution.

Phase Theory resolves the AMPS tension through a structural reinterpretation of all three principles. First, there is no horizon in the spacetime sense — only a saturation boundary. The saturation boundary is a topological interface in configuration space, not a null hypersurface in emergent spacetime. An infalling observer (a phase defect moving inward) crosses the saturation boundary and enters the saturated region. The observer does not encounter a localized concentration of outgoing energy quanta at the boundary, because the admissibility leakage is a distributed, collective process: the outgoing modes are delocalized outward-propagating wave modes, not a concentrated "wall" at the boundary.

Second, unitarity is preserved by GCE (Section 19). Information is not destroyed. Effective field theory breaks down near the saturation boundary — but not because of a firewall. It breaks down because the admissibility density is at its maximum, and the emergent field description (which is valid only for sub-maximal admissibility

densities) loses validity. This is a regime change, not a catastrophic physical event. The observer crossing the boundary enters a regime where the emergent description must be replaced by the phase-native description, but this does not correspond to any locally-experienced energy concentration.

Third, complementarity (the claim that infalling and exterior descriptions are consistent) holds in Phase Theory as a consequence of GCE: both descriptions are consistent coarse-grainings of the global phase configuration (Section 22). The AMPS argument assumed that these coarse-grainings are independent — that the information content of the exterior description and the interior description can be separately maximized. Phase Theory shows that this assumption is false: GCE enforces correlations between interior and exterior configurations, so the two descriptions are not independent, and the AMPS argument does not apply.

22. Complementarity Without Observers

Black hole complementarity (Susskind, Thorlacius, and Uglum 1993) was introduced as a resolution of the information paradox that preserves both unitarity and the equivalence principle. The proposal is that the infalling observer and the exterior observer have complementary, consistent descriptions of the same physical system, but that no single observer can access both descriptions simultaneously. This is analogous to wave-particle complementarity in quantum mechanics.

The proposal has been criticized on several grounds. It requires a non-standard "observer-dependent" ontology: what physically exists seems to depend on who is doing the observing. It relies on the assumption that the time for an infalling observer to reach the singularity is shorter than the scrambling time, which is a specific dynamical claim that is not always satisfied. And it is in tension with the AMPS argument (Section 21), which shows that the complementary descriptions cannot both be correct if effective field theory holds.

Phase Theory provides complementarity without any of these difficulties, and without reference to observers at all. The global phase configuration is a single,

observer-independent mathematical object $\varphi \in \text{Conf}_\phi$. Different observers in the emergent spacetime description access different coarse-grainings of φ . An infalling observer accesses the coarse-graining compatible with an admissibility-preserving inward path; an exterior observer accesses the coarse-graining compatible with the outgoing leakage modes. Both coarse-grainings are descriptions of the same global configuration; they are consistent by construction, because GCE enforces global consistency of φ at all times.

The reason no observer can simultaneously access both descriptions is not a statement about the limits of physical observation — it is a statement about the structure of the admissible configuration space. The inward path and the outward leakage pathways are topologically distinct families of paths in Conf_ϕ . A path that simultaneously follows both would exit the admissible configuration space (it would correspond to a phase field that simultaneously maximizes interior admissibility and maximizes exterior admissibility, which is forbidden by the saturation constraint). Complementarity is therefore a topological statement about the structure of Conf_ϕ , not a statement about the limits of observation.

23. Near-Extremal Phase-Saturated Objects

Extremal black holes are those that carry the maximum charge or spin compatible with their mass, corresponding to zero Hawking temperature. In general relativity, extremal Reissner-Nordström and extremal Kerr black holes have degenerate horizons (the inner and outer horizons coincide) and zero surface gravity. The standard derivation gives $T_H \rightarrow 0$ in the extremal limit.

In Phase Theory, extremal objects correspond to phase-saturated configurations in which the internal defect structure carries maximal topological charge relative to the total admissibility budget. Let Q_ϕ denote the total topological charge of the internal defect structure (the Phase Theory analogue of electric charge). The admissibility budget available for boundary winding is reduced by the self-energy of the topological charge:

$$A_{eff} = A_{sat} - Q_\phi^2 / (4\pi\rho_{max}) \quad (49.29)$$

As $Q_\phi \rightarrow Q_{max} = \sqrt{4\pi\rho_{max}A_{sat}}$, the effective boundary area $A_{eff} \rightarrow 0$, and the winding deficit approaches zero. Admissibility leakage vanishes: $T_\phi \rightarrow 0$. The saturation boundary becomes maximally rigid — there are no boundary admissibility fluctuations from which leakage modes can be generated.

Subleading corrections arise from quantum fluctuations of the defect topology. The defect charge Q_ϕ is quantized, and at finite temperature, topological charge-change events (monopole-antimonopole pair creation in the defect language) contribute a non-zero but exponentially suppressed leakage rate:

$$T_\phi(\text{near-extremal}) \sim \exp(-Q_\phi/\hbar) \quad (49.30)$$

This result is consistent with the known behavior of near-extremal black holes in string theory and supergravity, where quantum corrections to the extremal temperature are exponentially small. It is derived here from Phase Theory first principles without reference to strings, branes, or supergravity.

24. Rotating Phase-Saturated Objects (Kerr Analogue)

Phase-saturated objects carrying non-zero angular momentum correspond to rotating defect configurations. The phase field in a rotating saturated object has a helical winding structure: the phase winds not only radially (as in the non-rotating case) but also azimuthally, following the angular momentum of the rotating defect distribution. The saturation boundary geometry is oblate, with area:

$$A_{sat}(M, J) = 8\pi G(M^2 + M\sqrt{(M^2 - a^2c^2/G^2)})/c^4 \leq A_{sat}(M, 0) \quad (49.31)$$

where $a = J/(Mc)$ is the spin parameter. The inner and outer saturation radii are $r_{\pm} = GM/c^2 \pm \sqrt{G^2M^2/c^4 - a^2}$. The admissibility leakage for a rotating saturated object is anisotropic: leakage is preferentially along directions perpendicular to the rotation axis, where the helical winding structure is most tightly constrained. The leakage temperature is derived from the boundary entropy gradient for the oblate boundary geometry:

$$T_{\phi}(M, J) = \hbar(r_+ - r_-) / (4\pi k_B(r_+^2 + a^2)) \quad (49.32)$$

This is exactly the Kerr-Hawking temperature. The derivation follows the same path as Section 11, but with the oblate boundary area (49.31) substituted for the spherical area. No reference to the Kerr metric, the ergoregion, or the angular velocity of the horizon is required; the formula falls out of the boundary entropy calculation for a rotating phase field.

Superradiance in the Kerr case — the amplification of incident radiation by extracting rotational energy — has a natural Phase Theory analogue. An external phase perturbation incident on the oblate saturation boundary can, if its angular frequency satisfies $\omega < m\Omega_+$ (where Ω_+ is the rotation frequency of the outer boundary and m is the azimuthal mode number), couple to the helical winding structure and be amplified at the expense of the rotation energy. This is admissibility amplification: the incident perturbation increases the total outward admissibility leakage by coupling to the rotational winding reservoir. The amplification factor is calculable from the admissibility path integral and matches the standard Kerr superradiance result.

25. Charged Phase-Saturated Objects (Reissner-Nordström Analogue)

Phase-saturated objects with net topological charge correspond to objects carrying what appears, in the emergent description, as electric charge. The topological

charge Q_ϕ reduces the admissibility budget available for boundary leakage, as established in Section 23. The inner and outer saturation radii for a charged saturated object are:

$$r_\pm = GM/c^2 \pm \sqrt{(G^2 M^2/c^4 - GQ^2/(4\pi\epsilon_0 c^4))} \quad (49.33)$$

The boundary entropy for the charged case is computed from the effective boundary area $A_{\text{eff}} = 4\pi r_+^2$. The leakage temperature is:

$$T_\phi(M, Q) = \hbar c(r_+ - r_-) / (4\pi k_B r_+^2) \quad (49.34)$$

This is exactly the Reissner-Nordström Hawking temperature. The extremal limit $Q \rightarrow Q_{\text{max}}$ gives $r_+ \rightarrow r_-$ and therefore $T_\phi \rightarrow 0$, consistent with Section 23.

The charged case is particularly illuminating because it demonstrates the coupling between topological charge and admissibility leakage. The reduction in leakage temperature for a charged saturated object is not because the charge "blocks" the leakage pathways but because the charge self-energy reduces the effective admissibility budget at the boundary. The boundary has fewer available winding configurations (some are occupied by the charge structure), so the entropy is lower for the same mass, and therefore the temperature is lower. This is a concrete illustration of how the Phase Theory derivation subsumes the general relativistic result without importing GR machinery.

26. Phase-Native Derivation of the Stefan-Boltzmann Luminosity

Having established the thermal character and temperature of the admissibility leakage distribution, we now derive the total radiated power. The outgoing admissibility flux per unit area of the saturation boundary is determined by the

density of outgoing leakage modes per unit frequency times the energy per mode times the mode density of states.

The density of admissibility leakage modes per unit frequency per unit boundary area is determined by the counting of admissible outgoing configurations at frequency ω . By the same boundary entropy calculation as Section 12, the mode density is:

$$g(\omega) = \omega^2 / (\pi^2 c^3) \quad (49.35)$$

(summed over polarizations). This is the standard photon density of states in three dimensions — derived here from admissibility mode counting rather than from electromagnetic field quantization. The energy flux per unit area is:

$$j = \int_0^\infty \hbar \omega g(\omega) \langle |a_\omega|^2 \rangle d\omega = \int_0^\infty \hbar \omega^3 / (\pi^2 c^3) \cdot 1/(\exp(\hbar \omega / k_B T_\phi) - 1) d\omega \quad (49.36)$$

Evaluating the integral:

$$j = \pi^2 k_B^4 T_\phi^4 / (60 \hbar^3 c^2) = \sigma_\phi T_\phi^4 \quad (49.37)$$

where $\sigma_\phi = \pi^2 k_B^4 / (60 \hbar^3 c^2)$ is the Phase Theory Stefan–Boltzmann constant — identical to the standard Stefan–Boltzmann constant. The total luminosity is:

$$L = \sigma_\phi T_\phi^4 \cdot A_{sat} \quad (49.38)$$

Substituting $T_\phi = \hbar c^3 / (8\pi G M k_B)$ and $A_{sat} = 16\pi G^2 M^2 / c^4$:

$$dM/dt = -L/c^2 = -\hbar c^4 / (15360 \pi G^2 M^2) \quad (49.39)$$

confirming the evaporation constant $\alpha = \hbar c^4 / (15360\pi G^2)$ used in Section 18. The Phase Theory Stefan-Boltzmann constant is identical to the standard one because the admissibility mode counting (49.35) gives the same density of states as photon quantization. This reflects the deeper fact that in Phase Theory, photons *are* admissibility leakage modes — not an independent structure imposed from outside. The identification $\sigma_\phi = \sigma_{SB}$ is not a coincidence but a structural consequence of the identification of photons with phase modes established in Paper 14 of this series.

27. Phase Topology and Greybody Factors

The thermal spectrum derived in Section 26 describes the radiation emitted at the saturation boundary. The radiation actually detected by an asymptotic observer is modified by the propagation of modes from the boundary to infinity. In standard GR, this modification is described by greybody factors $\Gamma_l(\omega)$ — frequency-dependent transmission coefficients that arise from the effective potential barrier surrounding the black hole.

In Phase Theory, greybody factors arise from the admissibility structure of the leakage pathways at scales larger than the saturation boundary. A leakage mode propagating outward from the saturation boundary at frequency ω and angular momentum quantum number l must traverse a region of the phase substrate where the admissibility density varies from near-saturation (just outside the boundary) to negligible (far away). This admissibility variation acts as a potential barrier: modes with wavelength $\lambda \gg R_{sat}$ propagate through with transmission coefficient near unity (the admissibility density varies too slowly to scatter them), while modes with wavelength $\lambda \ll R_{sat}$ encounter significant admissibility topology barriers (the rapid variation in admissibility density acts as a scattering potential).

$$\Gamma_l(\omega) = \exp(-\Delta S_{barrier}(l, \omega)/k_B) \tag{49.40}$$

where $\Delta S_{barrier}$ is the admissibility entropy of the barrier region as a function of l and ω . For spherically symmetric saturated objects, this can be computed explicitly from

the admissibility density profile $\rho_{adm}(r)$ outside the saturation boundary. The result reproduces the standard greybody factors derived from the Regge–Wheeler potential in GR. The detailed calculation is deferred to Paper 50, but the functional form (49.40) and its limiting behaviors are established: $\Gamma_l(\omega) \rightarrow 1$ as $\omega \rightarrow \infty$, and $\Gamma_l(\omega) \rightarrow 0$ as $\omega \rightarrow 0$ for $l \geq 1$.

The important distinction from standard greybody factors is that in Phase Theory, the greybody factors encode topological information about the saturation boundary. For a boundary with non-trivial internal topology (charge, spin, or defect species composition), the admissibility density profile $\rho_{adm}(r)$ outside the boundary has topology-dependent features that produce resonances in $\Gamma_l(\omega)$ not present in classical GR calculations. These topology-dependent resonances constitute Prediction Class P49.5 (Section 36).

28. Black Hole Mining and Stimulated Leakage

Black hole mining (Unruh and Wald 1983, Brown et al. 2013) is the proposal that the rate of energy extraction from a black hole can be enhanced by external apparatus: a "mining bucket" lowered close to the horizon could collect Hawking radiation more efficiently than isotropic emission allows, in principle extracting energy faster than the natural evaporation rate. The proposal is theoretically contested but physically motivated.

Phase Theory predicts an analogous effect: stimulated admissibility leakage. An external phase perturbation — a coherent outgoing wave in the admissible configuration space — incident on the saturation boundary can couple to the leakage pathways and stimulate additional leakage at the same frequency. This is the Phase Theory analogue of stimulated emission: just as a photon incident on an excited atom can stimulate the emission of a second photon in the same mode, a leakage mode incident on the saturation boundary can stimulate the emission of additional leakage modes.

Proposition 49.2 (Stimulated Leakage Rate)

An external phase perturbation with amplitude A and frequency ω incident on the saturation boundary stimulates additional leakage at rate:

$$\Delta\Gamma(\omega) = A^2 \cdot \Gamma_0(\omega) \cdot (1 + \langle |a_\omega|^2 \rangle_{thermal})$$

(49.41)

where $\Gamma_0(\omega)$ is the spontaneous leakage rate at frequency ω and $\langle |a_\omega|^2 \rangle_{thermal}$ is the thermal occupation number (49.11). The factor $(1 + n)$ reflects the bosonic enhancement of stimulated over spontaneous emission.

This result has implications for protocols designed to extract energy from phase-saturated objects. An external phase antenna tuned to a frequency near the peak of the Planck distribution could, in principle, enhance the leakage rate by a factor of order $n_{thermal} + 1$. For macroscopic black holes, T_ϕ is extremely low, and $n_{thermal}$ is exponentially small at any laboratory frequency. But for near-Planckian remnants or analog gravity systems, stimulated leakage could be a significant effect. This provides an experimental handle for laboratory analog systems (Section 34) and constitutes a specific prediction distinguishing Phase Theory from both standard QFT and classical GR.

29. Comparison with Quantum Field Theory on Curved Spacetime

Feature	QFT on Curved Spacetime	Phase Theory (This Paper)
Origin of radiation	Bogoliubov transformation of vacuum modes; particle creation by curved spacetime	Admissibility leakage at saturation boundary; redistribution of phase configurations

Feature	QFT on Curved Spacetime	Phase Theory (This Paper)
Ontological basis	Quantum fields on classical curved background; vacuum fluctuations; virtual pairs	Global phase consistency enforcement; no fields, no vacuum, no particles at fundamental level
Trans-Planckian modes	Required; Hawking quanta exponentially blueshifted at early times; unresolved	Absent; hard UV cutoff at $\omega_{\max} \sim c/l_P$ from saturation constraint
Information	Radiation is exactly thermal; information appears lost; paradox unresolved	Radiation is thermal in marginal distribution; information encoded in GCE correlations; no paradox
Observer-dependence	Mode decomposition observer-dependent; infalling observer sees no radiation	Leakage rate is global scalar; all observers agree on total flux
Firewall	AMPS argument implies firewall if unitarity + EFT + equivalence principle all hold	No firewall; saturation boundary is smooth for infalling observer; no EFT required
Extremal limit	$T \rightarrow 0$; degeneracy of horizon; zero temperature limit of surface gravity	$T \rightarrow 0$ as winding deficit $\rightarrow 0$; exponentially suppressed subleading leakage
Unruh unification	Formally related via Rindler horizon; different derivations, same formula	Structurally unified as admissibility-slicing thermodynamics
Bekenstein-Hawking entropy	Gibbons-Hawking path integral; requires GR + analytic continuation	Derived from boundary admissibility count; requires only Axioms $\Phi 1$ - $\Phi 4$
Entropy derivation path	Euclidean path integral; area law as boundary term in GR action	Phase entropy gradient; area law from boundary admissibility topology

Table 29.1. Comparison of Quantum Field Theory on Curved Spacetime and Phase Theory for Hawking Radiation.

The comparison reveals a consistent pattern. Phase Theory reproduces every quantitative result of QFT on curved spacetime while resolving every paradox associated with that framework. The resolution is not achieved by modifying the numerical results but by providing a different explanatory foundation from which the same numerical results follow.

30. Comparison with String Theory Black Holes

Feature	String Theory	Phase Theory
Entropy derivation	D-brane microstate counting (Strominger & Vafa 1996); valid near extremality	Boundary admissibility counting; valid for all M
Radiation mechanism	String scattering and emission from D-brane bound states	Admissibility leakage at saturation boundary
Information	ER=EPR conjecture; wormholes connecting interior to radiation	GCE topological continuity; no wormholes required
Extremal limit	BPS bound; exact entropy at extremality from D-brane counting	Zero winding deficit; exponentially suppressed leakage
Foundational ingredients	10/11 dimensional spacetime; strings; branes; supersymmetry	Phase field on pre-geometric substrate; no additional dimensions; no SUSY
Predictive differences	Spectrum of string resonances in gravitational wave echoes	Topology-correlated mode correlations; phase relic mass distribution

Table 30.1. Comparison of String Theory and Phase Theory approaches to black hole thermodynamics.

String theory's most celebrated contribution to black hole physics is the microscopic derivation of the Bekenstein–Hawking entropy by Strominger and Vafa (1996) for extremal and near-extremal black holes. The derivation counts the number of D-brane microconfigurations with the correct charges and shows that it matches S_{BH} . Phase Theory's analogous derivation counts boundary admissibility configurations and achieves the same result for all values of mass, charge, and spin, not just near extremality.

31. Comparison with Loop Quantum Gravity Black Holes

Feature	Loop Quantum Gravity	Phase Theory
Entropy derivation	Spin-network edge states puncturing horizon; requires Immirzi parameter γ fixed by hand	Boundary admissibility configurations; no free parameters
Radiation mechanism	Quantum geometry transitions near quantum horizon	Admissibility leakage at saturation boundary
Trans-Planckian behavior	Discreteness of area spectrum provides natural UV cutoff	Saturation constraint provides hard UV cutoff; same effect, different origin
Singularity resolution	Loop quantum cosmology bounces; singularity replaced by maximum density state	Maximum admissibility density at saturation; no singularity
Falsifiable differences	Discrete area spectrum; modified dispersion relations; specific Immirzi parameter predictions	Phase relics; topology-correlated mode correlations; no Immirzi parameter

Table 31.1. Comparison of Loop Quantum Gravity and Phase Theory approaches to black hole thermodynamics.

Loop quantum gravity's derivation of black hole entropy (Rovelli 1996; Ashtekar, Baez, and Krasnov 1998) requires fixing the Immirzi parameter by demanding that the LQG entropy equal the Bekenstein–Hawking result. This is a one-parameter fine-tuning that has no independent justification within LQG. Phase Theory has no corresponding free parameter: the boundary admissibility count gives the correct entropy coefficient from first principles, determined entirely by the topological structure of the saturation boundary and the fundamental phase granularity l_P .

32. Comparison with Fuzzball and Firewall Proposals

Mathur's fuzzball proposal (Mathur 2009) replaces the classical black hole interior with a string-theoretic extended object: a "fuzzball" that fills the region bounded by what would classically be the event horizon. The fuzzball has no interior vacuum region; it is a quantum string-theoretic object, and information is encoded in the fuzzball microstate rather than in the Hawking radiation. The fuzzball radiates by a process analogous to thermal emission from the surface of an extended object, not by pair creation at a horizon.

Phase-saturated objects in Phase Theory share structural features with fuzzballs: both replace the classical black hole interior with a fundamentally different structure. Phase-saturated objects are not string-theoretic extended objects; they are regions of maximal phase admissibility density, bounded by a topological interface. Both frameworks agree that the classical notion of a black hole interior — a vacuum region bounded by an event horizon — is incorrect. Both predict that radiation is emitted from the boundary of the modified interior structure, not from pair creation in a vacuum.

The specific mechanisms and predictions differ importantly. Fuzzballs predict a near-horizon structure that is in principle detectable through gravitational wave echoes from black hole mergers: the fuzzball surface reflects gravitational waves, producing a series of echoes after the main merger signal. Phase-saturated objects make a similar but distinct prediction: the saturation boundary reflects gravitational-wave-like disturbances (propagating admissibility density perturbations), producing echoes with a specific frequency spacing determined by R_{sat} and the phase propagation speed. The echo timescale in Phase Theory is $\Delta t \sim 2R_{sat}/c = 4GM/c^3$, the same as for fuzzballs, but the echo amplitude decays according to the admissibility leakage rate rather than string-theoretic reflection coefficients, producing a distinguishable amplitude envelope.

The AMPS firewall proposal and Phase Theory's resolution have been compared in Section 21. In brief: the firewall is a symptom of the assumption that effective field theory holds at the horizon and that the descriptions of infalling and exterior observers are independent. Both assumptions are false in Phase Theory: effective field theory breaks down near the saturation boundary (because admissibility is at its maximum), and interior and exterior descriptions are correlated through GCE (so they are not independent). The firewall is dissolved, not by denying its logical derivation from the AMPS premises, but by rejecting the premises.

33. Primordial Phase-Saturated Objects and Cosmological Implications

Primordial black holes (PBHs) are hypothetical objects formed in the early universe from density perturbations during or shortly after inflation. They span an enormous mass range, from Planck mass to stellar mass and above, depending on the formation epoch. If PBHs exist, they are among the most important probes of Planck-scale physics: they are the only objects for which Hawking radiation might be observationally detectable (for PBHs of mass $M \sim 10^{15} \text{g}$, t_{evap} is of order the age of the universe).

Phase Theory modifies the standard PBH picture in three ways. First, PBH formation is governed by phase admissibility constraints during the early universe, which differ from those of the post-inflationary phase. In particular, during cosmological phase transitions in the phase substrate (events that appear as cosmological phase transitions in the effective spacetime description), the admissibility landscape changes: configurations that were admissible before the transition may be inadmissible after, and new saturation structures can form rapidly. This produces a modified PBH mass spectrum with characteristic peaks at masses corresponding to the admissibility transition scales.

Second, PBH evaporation products in Phase Theory include phase relics (Section 18) in addition to standard radiation. A population of phase relics with mass $M \sim M_P$ would

contribute to the dark matter density. The relic abundance depends on the initial PBH mass function and the evaporation history. For a monochromatic initial mass function at M_0 , the relic abundance is of order $\Omega_{\text{relic}} \sim (M_P/M_0) \cdot \Omega_{\text{PBH,initial}}$. The relic dark matter has a distinctive mass distribution peaked at M_P , distinguishable from other dark matter candidates.

Third, the evaporation spectrum of near-Planckian PBHs in Phase Theory deviates from the standard Hawking spectrum in a characteristic way: the spectrum has a hard UV cutoff at $\omega_{\text{max}} \sim c/l_P$ (Section 15) and topology-dependent resonances in the greybody factors (Section 27). The UV cutoff would modify the 511 keV annihilation signal from electron-positron pairs produced in PBH evaporation, and the topology-dependent resonances would produce frequency-dependent anisotropies in the diffuse gamma-ray background. Constraints from the CMB, gravitational wave background (from PBH mergers), and 21 cm observations provide complementary handles on the modified Phase Theory PBH population.

34. Analog Gravity and Laboratory Tests

Analog gravity systems — physical systems that exhibit mathematical parallels with curved spacetime — provide laboratory probes of Hawking-like radiation. The best-studied examples are sonic black holes (Unruh 1981; Weinfurtner et al. 2011), where the speed of sound plays the role of the speed of light and a supersonic flow creates an acoustic "event horizon"; optical analogues using ultrashort laser pulses; and Bose-Einstein condensate (BEC) horizons (Steinhauer 2016), where the speed of sound in the condensate defines the horizon. Hawking-like radiation has been observed in BEC systems (Steinhauer 2016, Muñoz de Nova et al. 2019).

Phase Theory makes specific predictions for analog systems that differ from standard QFT predictions. First, the analog Hawking spectrum should be exactly thermal, not approximately thermal. The Phase Theory argument for thermality (Section 10) relies on the tracing over inaccessible interior configurations — a structural argument that applies to any system with a saturation-type boundary, not just gravitational black holes. Analog systems with sharp "saturation boundaries"

(regions of maximum information density in the effective description) should exhibit exact Planck distributions, while systems with soft boundaries should exhibit deviations. Second, correlations between ingoing and outgoing modes in the analog system should encode topological information about the boundary structure. In particular, the two-point function of outgoing mode amplitudes at different frequencies should exhibit non-trivial correlations arising from the topology of the effective saturation boundary. This is a measurable prediction: BEC systems can be probed with sufficient precision to detect these correlations.

Third, the trans-Planckian problem should not appear in well-controlled analog systems. In BEC systems, the analog of the Planck length is the healing length ξ of the condensate. The Phase Theory prediction is that modes with wavelength below ξ do not participate in the analog Hawking effect at all — there is a hard cutoff in the analog spectrum at $\omega \sim c_s/\xi$ (where c_s is the speed of sound). This is precisely what is observed in numerical simulations of BEC analog gravity systems, consistent with Phase Theory and inconsistent with naive field-theoretic predictions that include trans-Planckian modes.

35. Objections and Responses

(O1) "Phase Theory is not yet a complete quantum theory — how can you derive radiation rigorously?"

The admissibility framework is defined axiomatically in Axioms $\Phi 1$ – $\Phi 4$. Derivations in this paper follow from these axioms through well-defined mathematical steps: the admissibility density functional (Definition 49.3), the GCE operator (Definition 49.2), the boundary entropy formula (equation 49.12), and the thermodynamic identity $T = (\partial S / \partial E)^{-1}$. Rigor does not require that Phase Theory be a quantization of some pre-existing classical theory; it requires only that the axioms be stated precisely and the derivations follow from them. Phase Theory is a self-contained axiomatic framework, and the derivations herein are rigorous within that framework. The question of whether Phase Theory has a canonical quantization formulation is a separate question addressed in Paper 50.

(O2) "You have reproduced Hawking's results — why should we prefer Phase Theory's derivation?"

Reproduction of known results is necessary but not sufficient. Any acceptable derivation must reproduce (49.1) and (49.2); Phase Theory does. What Phase Theory additionally provides is: (a) a derivation free from trans-Planckian modes, (b) a resolution of the information paradox without additional postulates, (c) a resolution of the firewall paradox, (d) a structural unification of the Hawking and Unruh effects, (e) five classes of falsifiable predictions distinguishing Phase Theory from QFT (Section 36). A derivation that achieves all of (a)–(e) while reproducing all known results is strictly preferable to one that achieves only the reproduction.

(O3) "The winding deficit and admissibility density are not independently defined — the derivation is circular."

The winding deficit (Definition 49.4) is defined as the integral of the difference between interior and exterior winding number densities; it depends on the global topology of the phase field. The admissibility density (Definition 49.3) is defined as the functional derivative of log-admissibility with respect to the local phase field; it is a local functional. These definitions are mathematically independent: neither is defined in terms of the other. The relationship between them — that a high winding deficit implies a high boundary admissibility gradient — is a theorem (Theorem 49.3), not a circular definition.

(O4) "The information paradox is not 'dissolved' — it requires a concrete mechanism for information transfer."

GCE provides this mechanism explicitly. Information about the initial state is encoded in the global phase configuration $\varphi \in \text{Conf}_\phi$. As evaporation proceeds, GCE enforces consistency between the evolving interior configuration and the outgoing leakage modes. The correlations between sequential outgoing modes, mediated by GCE, encode the initial state information in the final radiation field. The mechanism is: GCE-enforced correlations in the leakage pathway amplitudes a_k carry information from the interior to the exterior. This is a concrete mechanism, not

merely an assertion. The calculation of these correlations is deferred to Paper 50 (two-point functions of outgoing modes).

(O5) "Phase Theory predicts the same temperature as QFT — it has no additional predictive content."

This objection is incorrect. Section 36 enumerates five prediction classes with signatures distinguishable from QFT: a hard UV spectral cutoff (P49.1), topology-correlated mode correlations (P49.2), near-Planckian temperature deviations (P49.3), phase relics as dark matter (P49.4), and topology-dependent greybody resonances (P49.5). None of these is predicted by standard QFT on curved spacetime. The temperature formula matches by design: any correct theory must reproduce (49.1). The distinctions appear in the corrections and correlations, not in the leading-order temperature.

(O6) "The comparison with GR horizon thermodynamics is merely notational."

The derivation path is structurally different from GR horizon thermodynamics. GR computes entropy from the Gibbons–Hawking boundary term in the Euclidean path integral; Phase Theory computes it from the admissibility microstate count at the saturation boundary. GR computes temperature from the surface gravity of the event horizon; Phase Theory computes it from the thermodynamic identity applied to the phase entropy. The mathematical results agree (as they must, if Phase Theory is correct), but the conceptual content is entirely different. The matching of results is a consistency check, confirming that Phase Theory's axioms reproduce GR results in the appropriate limit. It is not a redefinition: the Phase Theory derivation makes additional predictions (Section 36) that go beyond GR.

36. Five Classes of Falsifiable Predictions

Prediction Class P49.1 — Hard Ultraviolet Spectral Cutoff

The emission spectrum of any phase-saturated object must have a hard UV cutoff at $\omega_{max} = c/l_P \sim 1.86 \times 10^{43}$ rad/s, unlike the standard Hawking spectrum which has a power-law tail extending to arbitrarily high frequencies. *Signature:* In analog systems, a spectral cutoff at the analog Planck frequency $\omega_{analog} = c/s\xi$. *Protocol:* BEC analog systems with well-characterized healing length should show a sharp cutoff in the analog Hawking spectrum. Gravitational PBH evaporation data (from future detectors sensitive to near-Planckian PBH evaporation) should show a hard cutoff in the high-energy gamma-ray spectrum rather than the power-law tail predicted by QFT.

Prediction Class P49.2 — Topology-Correlated Mode Correlations

The two-point function of outgoing leakage mode amplitudes $\langle a_k a_{k'}^* \rangle$ should encode the winding topology of the initial saturated object in a non-trivial way. Specifically, modes at frequencies related by the saturation boundary's topological resonances should show correlated amplitudes, while modes at generic frequencies should appear independent (thermal). *Signature:* In BEC systems, frequency correlations in the Hawking radiation spectrum that encode the topology of the effective horizon. *Protocol:* Two-point measurements of analog Hawking spectra in BEC systems with controlled effective topological structure, compared with topologically trivial control systems.

Prediction Class P49.3 — Near-Planckian Temperature Deviations

For phase-saturated objects with M approaching M_P , the temperature deviates from T_H by corrections of order $O(l_P/R_{sat}) = O(M_P/M)$:

$$T_\phi = T_H(1 + c_1 M_P/M + c_2(M_P/M)^2 + \dots)$$

(49.42)

where c_1, c_2 are calculable from Phase Theory (Paper 50). QFT predicts no such correction (the standard formula remains valid down to the Planck mass in the absence of a separate UV completion). *Protocol:* Near-Planckian PBH evaporation spectrum; requires sensitivity to gamma-ray fluxes from PBHs with current mass near M_P .

Prediction Class P49.4 — Phase Relics as Dark Matter

Black hole evaporation does not terminate in complete evaporation but in a stable or long-lived phase relic of mass $M_{\text{relic}} \sim M_P$ with a distinctive topological charge structure. These relics contribute to the dark matter density with a mass distribution sharply peaked at M_P (unlike WIMPs, axions, or other dark matter candidates). *Signature:* A component of dark matter with Planck-mass particles, interacting only gravitationally and through topological charge. *Protocol:* Searches for Planck-mass particles in cosmic ray detectors; gravitational microlensing surveys sensitive to Planck-mass objects; modified dark matter power spectrum from Planck-mass thermal relics in early universe.

Prediction Class P49.5 — Topology-Dependent Greybody Resonances

Greybody factors in Phase Theory exhibit topology-dependent resonances not present in classical QFT calculations. For a saturated object with non-trivial internal topology (charge Q , spin J , or specific defect species composition), the greybody factor $\Gamma_l(\omega)$ has narrow resonances at frequencies ω_n determined by the topological structure of the saturation boundary. *Signature:* Quasi-normal

mode spectrum of black holes with narrow additional resonances superposed on the standard QNM spectrum. *Protocol:* Gravitational wave observations of black hole quasi-normal modes with high signal-to-noise ratio (LISA, Einstein Telescope, Cosmic Explorer) to resolve resonance fine structure not predicted by GR.

37. Relation to the Thermodynamics of Spacetime

Jacobson (1995) demonstrated that the Einstein field equations can be derived from the thermodynamic identity $\delta Q = T \delta S$ applied to local causal horizons, where the entropy is proportional to the horizon area and the heat flux is the matter energy flux through the horizon. This elegant result suggests that gravity is not a fundamental force but an emergent thermodynamic phenomenon. Padmanabhan (2010, 2013) and others have extended this insight, showing that the gravitational dynamics can be reformulated as a thermodynamic extremization principle, and that the holographic degrees of freedom on surfaces play the primary role.

Phase Theory provides a deeper foundation for these results. In Phase Theory, the thermodynamics of spacetime is not a coincidence or a mathematical curiosity — it is a structural consequence of the admissibility framework. The emergent metric tensor (Axiom $\Phi 3$) is determined by the two-point correlations of the phase field, which are themselves governed by the admissibility constraints. The Einstein field equations emerge as the consistency conditions that the emergent metric must satisfy, given the admissibility constraints on the phase field. This is the Phase Theory derivation of GR (Papers 31–35).

The Jacobson result follows from Phase Theory as a limiting case. The local causal horizon to which Jacobson applies the thermodynamic identity is the Phase Theory admissibility boundary for a small region of phase substrate. The entropy is the

boundary admissibility entropy (equation 49.12). The temperature is the admissibility leakage temperature (equation 49.17) for the local causal horizon. The heat flux is the admissibility flux through the boundary (the leakage rate). The thermodynamic identity $\delta Q = T\delta S$ then becomes, in the Phase Theory language, the GCE conservation law (total admissibility is conserved). The Raychaudhuri equation and the Clausius relation that Jacobson uses correspond, in Phase Theory, to the equations governing the evolution of the admissibility density distribution near a local phase boundary.

Implications for entropic gravity (Verlinde 2011) are also immediate. Verlinde's proposal that gravity is an entropic force arising from the thermodynamics of information on holographic screens is, in Phase Theory, a consequence of admissibility leakage at boundaries of varying admissibility density. The gravitational force is the macroscopic emergent description of the tendency of the phase field to evolve toward configurations of maximum global admissibility. The entropic force formula $F = T \partial S / \partial x$ is the Phase Theory admissibility gradient force, derived in Papers 32–33.

38. Holography and AdS/CFT from Phase Theory

The AdS/CFT correspondence (Maldacena 1997) posits a duality between string theory in $AdS_5 \times S^5$ and $N = 4$ supersymmetric Yang–Mills theory on the four-dimensional conformal boundary. This duality is one of the most important results in theoretical physics of the past three decades, providing a concrete realization of the holographic principle and a computational tool for strongly-coupled quantum field theories.

Phase Theory provides a derivation of holography from first principles. The bulk-boundary duality arises from the relationship between interior admissibility structure and boundary phase modes. The saturation boundary of a phase-saturated object is a boundary in exactly this sense: the interior admissibility structure is

encoded in the boundary configuration, because GCE enforces global consistency between interior and boundary admissibility. The boundary information content is exactly sufficient to specify the interior, because the interior is saturated (its admissibility is at maximum) and therefore its configuration is uniquely determined by the boundary conditions.

More precisely: a saturated interior has $\Omega_{\text{interior}} = 1$ (all interior microstates are identical — the interior is frozen). The information about the initial configuration that led to this saturated state is therefore entirely encoded in the boundary. The boundary admissibility configurations (counted by $|\Omega_{\text{boundary}}|$) encode this information. The one-to-one map between interior configurations and boundary configurations is the holographic map. In the effective spacetime description, this map is the AdS/CFT duality.

Hawking radiation is a finite-temperature manifestation of this holographic encoding. At zero temperature, the saturation boundary is static: no leakage occurs, and the boundary configuration is fixed. At finite temperature (i.e., when the saturated region has non-zero admissibility leakage), the boundary configuration evolves over time as leakage pathways carry admissibility outward. This evolution corresponds, in the AdS/CFT language, to the thermal field theory on the boundary at temperature T_ϕ . The Hartle–Hawking state (Hartle and Hawking 1983), the Unruh state, and the Boulware state of standard black hole physics are, in Phase Theory, different admissibility slicings of the global phase configuration, corresponding to different thermal states of the boundary phase theory.

The ER=EPR conjecture (Maldacena and Susskind 2013) — that entangled black holes are connected by Einstein–Rosen bridges (wormholes) — has a natural Phase Theory interpretation. Entangled phase-saturated objects are connected through GCE: the admissibility constraints enforce correlations between their boundary configurations. This GCE-mediated correlation is the Phase Theory counterpart of the wormhole connection. No actual spacetime wormhole is required; the "connection" is a correlation in the global admissible configuration space, mediated by GCE. This provides a cleaner ontological basis for ER=EPR while preserving all its computational and conceptual content.

39. Summary and Outlook

This paper has derived Hawking radiation in Phase Theory without invoking quantum fields, vacuum fluctuations, particle-antiparticle pairs, or curved-spacetime field quantization. The derivation proceeds from four axioms ($\Phi 1$ – $\Phi 4$) through the following logical chain: phase saturation (Definition 49.1) $\rightarrow$ winding deficit (Definition 49.4, Proposition 49.1) $\rightarrow$ admissibility leakage (Theorem 49.3, Definition 49.5) $\rightarrow$ thermality of outgoing modes (Theorem 49.4) $\rightarrow$ phase temperature (equation 49.17) $\rightarrow$ Bekenstein-Hawking entropy (equation 49.18) $\rightarrow$ energy conservation (Theorem 49.6) $\rightarrow$ evaporation (equation 49.25). Every step is forced by the axioms; no additional assumptions are introduced.

The main results established in this paper are: (i) the Hawking temperature $T_\phi = \hbar c^3 / (8\pi G M k_B)$ is derived from the thermodynamic identity applied to the phase entropy at saturation, without surface gravity, Bogoliubov transformations, or mode decompositions (Section 11); (ii) the Bekenstein-Hawking entropy $S_{BH} = k_B A / (4l_P^2)$ is derived from boundary admissibility counting, without GR path integrals, D-brane counting, or Immirzi parameters (Section 12); (iii) the information paradox is dissolved by GCE-enforced topological continuity of the configuration space (Section 19); (iv) the Page curve is recovered from admissibility topology without quantum extremal surfaces (Section 20); (v) the firewall paradox is resolved without exotic boundary conditions (Section 21); (vi) the trans-Planckian problem does not arise, because admissibility saturation imposes a hard UV cutoff (Section 15); (vii) the Unruh and Hawking effects are structurally unified as admissibility-slicing thermodynamics (Section 17); (viii) the Kerr, Reissner-Nordström, and extremal cases are derived as structural generalizations (Sections 23–25).

What remains to be done is substantial. Paper 50 will compute the two-point functions of outgoing leakage modes explicitly, providing quantitative predictions for Prediction Classes P49.2 and P49.3. Paper 51 will develop the angular momentum superradiance analogue in Phase Theory in full detail. Paper 52 will derive the primordial phase-saturated object mass spectrum from Phase Theory's description of early-universe admissibility transitions, providing quantitative

predictions for Prediction Class P49.4. Paper 53 will develop explicit experimental protocols for laboratory analog gravity tests, providing quantitative predictions for Prediction Classes P49.1 and P49.5. Paper 54 will address the Phase Theory description of cosmological horizons — the de Sitter horizon and the particle horizon — and derive their thermodynamic properties from admissibility principles.

The philosophical implications of this derivation deserve emphasis. In Phase Theory, black holes are not regions of extreme spacetime curvature from which gravity prevents escape. They are regions in which phase has reached its maximum admissible density — regions of maximum information density in the fundamental substrate. Hawking radiation is not a quantum gravitational effect superimposed on a classical geometry. It is the inevitable consequence of global phase consistency applied to a maximally saturated local region: the phase substrate cannot maintain global admissibility without leaking outward. The radiation was always required. The challenge for future work is to make this requirement precise enough to be falsified — or confirmed — by observation.

Black holes are not regions from which nothing escapes. They are regions in which phase has reached its maximum admissible density — and from which, therefore, phase reconfiguration must leak. Hawking radiation is not a quantum effect superimposed on a classical geometry. It is the inevitable consequence of global phase consistency applied to a maximally saturated local region. Phase Theory does not add a mechanism to explain Hawking radiation. It reveals that radiation was always required.

References

[1] Hawking, S. W. (1974). Black hole explosions? *Nature*, 248(5443), 30–31.

- [2] Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics*, 43(3), 199–220.
- [3] Hawking, S. W. (1976). Breakdown of predictability in gravitational collapse. *Physical Review D*, 14(10), 2460–2473.
- [4] Bekenstein, J. D. (1973). Black holes and entropy. *Physical Review D*, 7(8), 2333–2346.
- [5] Bekenstein, J. D. (1972). Black holes and the second law. *Lettere al Nuovo Cimento*, 4(15), 737–740.
- [6] Unruh, W. G. (1976). Notes on black-hole evaporation. *Physical Review D*, 14(4), 870–892.
- [7] Unruh, W. G. (1981). Experimental black-hole evaporation? *Physical Review Letters*, 46(21), 1351–1353.
- [8] Unruh, W. G., & Wald, R. M. (1983). How to mine energy from a black hole. *General Relativity and Gravitation*, 15(3), 195–199.
- [9] Page, D. N. (1993). Information in black hole radiation. *Physical Review Letters*, 71(23), 3743–3746.
- [10] Almheiri, A., Marolf, D., Polchinski, J., & Sully, J. (2013). Black holes: complementarity or firewalls? *Journal of High Energy Physics*, 2013(2), 62.
- [11] Penington, G. (2020). Entanglement wedge reconstruction and the information paradox. *Journal of High Energy Physics*, 2020(9), 2.
- [12] Almheiri, A., Mahajan, R., Maldacena, J., & Zhao, Y. (2020). The Page curve of Hawking radiation from semiclassical geometry. *Journal of High Energy Physics*, 2020(3), 149.
- [13] Almheiri, A., Engelhardt, N., Marolf, D., & Maxfield, H. (2020). Entanglement island from holography. *Journal of High Energy Physics*, 2020(9), 1.
- [14] Mathur, S. D. (2009). The information paradox: a pedagogical introduction. *Classical and Quantum Gravity*, 26(22), 224001.
- [15] Susskind, L., Thorlacius, L., & Uglum, J. (1993). The stretched horizon and black hole complementarity. *Physical Review D*, 48(8), 3743–3761.
- [16] Maldacena, J. M. (1997). The large N limit of superconformal field theories and supergravity. *International Journal of Theoretical Physics*, 38(4), 1113–1133.
- [17] Maldacena, J., & Susskind, L. (2013). Cool horizons for entangled black holes. *Fortschritte der Physik*, 61(9), 781–811.
- [18] Wald, R. M. (1994). *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. University of Chicago Press.

- [19] Birrell, N. D., & Davies, P. C. W. (1982). *Quantum Fields in Curved Space*. Cambridge University Press.
- [20] Parker, L. (1969). Quantized fields and particle creation in expanding universes. *Physical Review*, 183(5), 1057–1068.
- [21] Israel, W. (1976). Thermo-field dynamics of black holes. *Physics Letters A*, 57(2), 107–110.
- [22] Hartle, J. B., & Hawking, S. W. (1983). Wave function of the universe. *Physical Review D*, 28(12), 2960–2975.
- [23] Gibbons, G. W., & Hawking, S. W. (1977). Action integrals and partition functions in quantum gravity. *Physical Review D*, 15(10), 2752–2756.
- [24] Strominger, A., & Vafa, C. (1996). Microscopic origin of the Bekenstein-Hawking entropy. *Physics Letters B*, 379(1–4), 99–104.
- [25] Rovelli, C. (1996). Black hole entropy from loop quantum gravity. *Physical Review Letters*, 77(16), 3288–3291.
- [26] Ashtekar, A., Baez, J., & Krasnov, K. (1998). Quantum geometry of isolated horizons and black hole entropy. *Advances in Theoretical and Mathematical Physics*, 4(1), 1–94.
- [27] Jacobson, T. (1995). Thermodynamics of spacetime: the Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263.
- [28] Padmanabhan, T. (2010). Thermodynamical aspects of gravity: new insights. *Reports on Progress in Physics*, 73(4), 046901.
- [29] Verlinde, E. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011(4), 29.
- [30] Polchinski, J. (1995). Dirichlet branes and Ramond-Ramond charges. *Physical Review Letters*, 75(26), 4724–4727.
- [31] Steinhauer, J. (2016). Observation of quantum Hawking radiation and its entanglement in an analogue black hole. *Nature Physics*, 12(10), 959–965.
- [32] Muñoz de Nova, J. R., et al. (2019). Observation of thermal Hawking radiation and its temperature in an analogue black hole. *Nature*, 569(7758), 688–691.
- [33] Weinfurter, S., Tedford, E. W., Penrice, M. C. J., Unruh, W. G., & Lawrence, G. A. (2011). Measurement of stimulated Hawking emission in an analogue system. *Physical Review Letters*, 106(2), 021302.
- [34] Hawking, S. W., Perry, M. J., & Strominger, A. (2016). Soft hair on black holes. *Physical Review Letters*, 116(23), 231301.

- [35] Hanks, M. (2026). Phase Theory: The Primacy of Phase as Ontological Foundation. *Phase Theory Series*, Paper 1. Preprint (2026).
- [36] Hanks, M. (2026). Phase Saturation Boundaries I: Definition, Properties, and Dynamic Onset. *Phase Theory Series*, Paper 47. Preprint (2026).
- [37] Hanks, M. (2026). Phase Saturation Boundaries II: Topological Classification and Boundary Entropy. *Phase Theory Series*, Paper 48. Preprint (2026).
- [38] Corley, S., & Jacobson, T. (1996). Hawking spectrum and high frequency dispersion. *Physical Review D*, 54(2), 1568–1586.
- [39] Penrose, R. (1965). Gravitational collapse and space-time singularities. *Physical Review Letters*, 14(3), 57–59.
- [40] Brown, A. R., Dahlen, S., & Singh, A. (2013). Revisiting the black hole mining process. *Physical Review D*, 88(8), 084023.